

Delta-dependent Notch activation closes the early neuroblast temporal program to promote lineage progression and neurogenesis termination in *Drosophila*

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Abstract

Summary

Neuroblasts in *Drosophila* divide asymmetrically, sequentially expressing a series of intrinsic factors to generate a diversity of neuron types. These intrinsic factors known as temporal factors dictate timing of neuroblast transitions in response to steroid hormone signaling and specify early versus late temporal fates in neuroblast neuron progeny. After completing their temporal programs, neuroblasts differentiate or die, finalizing both neuron number and type within each neuroblast lineage. From a screen aimed at identifying genes required to terminate neuroblast divisions, we identified Notch and Notch pathway components. When Notch is knocked down, neuroblasts maintain early temporal factor expression longer, delay late temporal factor expression, and continue dividing into adulthood. We find that Delta, expressed in cortex glia, neuroblasts, and after division, their GMC progeny, regulates neuroblast Notch activity. We also find that Delta in neuroblasts is expressed high early, low late, and is controlled by the intrinsic temporal program: early factor Imp promotes Delta, late factors Syp/E93 reduce Delta. Thus, in addition to systemic steroid hormone cues, forward lineage progression is controlled by local cell-cell signaling between neuroblasts and their cortex glia/GMC neighbors: Delta transactivates Notch in neuroblasts bringing the early temporal program and early temporal factor expression to a close.

eLife assessment

This **useful** study reports on how Notch activity regulates the termination of neurogenesis in central brain during larval-pupal stages in *Drosophila*. The evidence supporting the claims is **solid**. The work will be of interest to developmental neurobiologists.

Introduction

In most metazoans, termination of neurogenesis is an essential part of organism development, ensuring the formation of functional neural circuits and an adult brain of proper size and structure. Prolonged or ectopic neurogenesis can lead to cortical malformations and has been linked to neurodevelopmental disorders, including autism (Hazlett et al., 2011 [DOI](#); Marchetto et al., 2017 [DOI](#); Patzlaff et al., 2018 [DOI](#); Casingal et al., 2020 [DOI](#); Ossola and Kalebic, 2022 [DOI](#)). While the vast majority of neurons are generated during development, it still remains unclear how neurogenesis becomes progressively restricted and, in most cases, ends altogether after development is completed.

We use the genetically tractable model organism, *Drosophila melanogaster*, to determine how extrinsic cues, local and systemic, integrate with neural stem cell intrinsic cues to control neurogenesis timing and termination during development. The *Drosophila* CNS consists of two bilaterally symmetric brain hemispheres and a ventral nerve cord that is functionally equivalent to the mammalian spinal cord. Each brain hemisphere contains an optic lobe and an equally sized central brain (CB) region that harbors distinctive neuropils for information processing. Neurons in the CB region are generated during development from the asymmetric cell divisions of a defined number of neural stem cells known as neuroblasts (NBs) in *Drosophila*. NBs in the CB region (referred to as CB NBs) are specified during embryogenesis and undergo stereotypic patterns of cell division (Truman and Bate, 1988 [DOI](#); Ito and Hotta, 1992 [DOI](#); Doe, 2008 [DOI](#); Siegrist et al., 2010 [DOI](#); Homem and Knoblich, 2012 [DOI](#)). Except for the mushroom body (MB) NB subset, all CB NBs enter and exit quiescence during the embryonic to larval transition and terminally differentiate or die four to five days later during early pupal stages (Truman and Bate, 1988 [DOI](#); Ito and Hotta, 1992 [DOI](#); Doe, 2008 [DOI](#); Maurange et al., 2008 [DOI](#); Siegrist et al., 2010 [DOI](#); Yang et al., 2017 [DOI](#)).

Once CB NBs reactivate from quiescence in response to dietary nutrients, they divide continuously while changing gene expression over time (Britton and Edgar, 1998 [DOI](#); Chell and Brand, 2010 [DOI](#); Sousa-Nunes et al., 2011 [DOI](#); Liu et al., 2015 [DOI](#); Syed et al., 2017 [DOI](#); Yuan et al., 2020 [DOI](#)). These controlled transitions of gene expression over time, referred to as temporal patterning, allow for a restricted set of neural stem cells to generate a pool of molecularly and functionally diverse neuron types (Isshiki et al., 2001 [DOI](#); Maurange et al., 2008 [DOI](#); Bayraktar and Doe, 2013 [DOI](#); Liu et al., 2015 [DOI](#); Bahrapour et al., 2017 [DOI](#); Ren et al., 2017 [DOI](#); Syed et al., 2017 [DOI](#); Miyares and Lee, 2019 [DOI](#)). Early larval temporal factors include the Zinc finger transcription factor, Castor (Cas), the orphan nuclear receptor, Seven-up (Svp), the RNA binding protein, IGF-II mRNA-binding protein (Imp), as well as others (Maurange et al., 2008 [DOI](#); Liu et al., 2015 [DOI](#); Ren et al., 2017 [DOI](#); Syed et al., 2017 [DOI](#)). Svp expression primes NBs to respond to a systemic pulse of steroid hormone (ecdysone) during larval stages and switch temporal factor expression from early to late (Ren et al., 2017 [DOI](#); Syed et al., 2017 [DOI](#)). Late temporal factors include the RNA binding protein, Syncrin (Syp), the steroid hormone-induced transcription factor, Eip93F (E93), as well as others (Liu et al., 2015 [DOI](#); Syed et al., 2017 [DOI](#); Pahl et al., 2019 [DOI](#)). Imp (early) and Syp (late) mutually inhibit each other and are expressed in opposing gradients in NBs (Liu et al., 2015 [DOI](#); Yang et al., 2017 [DOI](#)). Imp keeps CB NBs “young” by inhibiting Syp and Mediator complex activity, whereas Syp inhibits Imp and promotes nuclear accumulation of the pro-differentiation transcription factor Prospero (Pros) in most CB NBs (Homem et al., 2014 [DOI](#); Liu et al., 2015 [DOI](#); Yang et al., 2017 [DOI](#)). During early pupal stages, CB NBs undergo reductive divisions and terminally differentiate, except for the MB NB subset, which divides several days longer and undergo autophagy/apoptosis prior to adult eclosion (Maurange et al., 2008 [DOI](#); Siegrist et al., 2010 [DOI](#); Homem et al., 2014 [DOI](#); Pahl et al., 2019 [DOI](#)). Independent of neurogenesis timing and the mechanism by which CB NB stop divisions, temporal patterning plays a key role in controlling numbers and types of neurons made within each of the NB lineages (Maurange et al., 2008 [DOI](#); Tsuji et al., 2008 [DOI](#); Bahrapour et al., 2017 [DOI](#); Yang et al., 2017 [DOI](#); Pahl et al., 2019 [DOI](#)).

From a targeted RNAi screen aimed at identifying genes required to terminate CB NB divisions and neurogenesis, we identified Notch and Notch pathway components. Notch is an evolutionarily conserved cell-cell signaling pathway classically known for regulating binary cell fate decisions, “A” versus “B” (Muskavitch, 1994; Cau and Blader, 2009). Here we show that Notch signaling also regulates binary temporal decisions, “early” versus “late”. In *Drosophila*, there is one Notch receptor and two ligands, Delta (Dl) and Serrate (Ser). Notch receptor is proteolytically cleaved after ligand binding, first by Kuzbanian (Kuz), an ADAM metalloprotease, and then by γ -secretase. Cleaved Notch ICD (intracellular domain) relocates to the nucleus where it binds to Suppressor of Hairless [Su(H)] and Mastermind to regulate gene expression (Rebay et al., 1991; Fortini and Artavanis-Tsakonas, 1994; Pan and Rubin, 1997; Strooper et al., 1999; Mumm et al., 2000; Kitagawa et al., 2001; Kopan and Ilagan, 2009). We recently reported that Notch signaling regulates CB NB quiescence during the embryonic to larval transition (Sood et al., 2022). When Notch is knocked down, some CB NBs continue dividing during this transition. We also reported that Notch activity becomes attenuated in quiescent CB NBs because CB NBs are no longer dividing and producing Delta-expressing GMC daughters for Notch pathway transactivation. Moreover, low Notch is necessary for CB NBs to reactivate from quiescence in response to dietary nutrients (Sood et al., 2022).

Here we report that Notch signaling also regulates neurogenesis termination during pupal stages. When Notch is knocked down, CB NBs maintain early temporal factor expression longer resulting in a delay of late temporal factor expression with prolonged neurogenesis into late pupal stages and early adulthood. This defect in temporal patterning (switching from early to late) occurs well after CB NB exit from quiescence suggesting that Notch is required at multiple times throughout development in controlling CB NB proliferation decisions.

Furthermore, we determine that Delta is the Notch ligand that activates Notch in CB NBs and reductions in Delta also lead to defects in CB NB temporal patterning. Moreover, we find that Delta in CB NBs, which is segregated to GMCs after cell division to transactivate Notch is regulated by CB NB temporal factors. Early factor Imp promotes Delta, whereas late factors Syp and E93 reduce Delta. Together, we report that Notch signaling positively regulates forward lineage progression by closing off the early temporal window and control of Notch pathway activity is regulated by CB NB intrinsic temporal factors.

Results

Notch signaling is required for CB NB elimination and termination of neurogenesis

All central brain neuroblasts (CB NBs), except the MB NB subset (four per brain hemisphere) terminally differentiate or die during early pupal stages (Fig. 1A) (Truman and Bate, 1988; Ito and Hotta, 1992; Maurange et al., 2008; Siegrist et al., 2010; Homem et al., 2014; Yang et al., 2017). MB NBs divide several days longer and undergo apoptotic/autophagic cell death shortly before adult eclosion (Fig. 1A,B) (Siegrist et al., 2010; Pahl et al., 2019). No CB NBs remain in adult animals and no new neurons are produced (Fig. 1A,B) (Truman and Bate, 1988; Siegrist et al., 2010; Yang et al., 2017). From a targeted RNAi screen aimed at identifying genes required to terminate NB divisions and neurogenesis, we identified Notch (N) and Notch pathway components. At 48 hours APF (after pupal formation), midway through pupal stages, control animals have only the four MB NBs remaining in each brain hemisphere (Fig. 1A,B). In contrast when Notch was knocked down in NBs (*worGAL4,UAS-N RNAi #HMS00001*), on average five additional CB NBs remained (Fig. 1C,D). In one-day old *N RNAi* adults, CB NBs were also present, but not the MB NBs (Fig. 1C,D). Ectopically persisting *N RNAi* CB NBs (CB NBs at 48 hours APF and beyond) expressed the NB transcription factor Deadpan (Dpn), the S-phase indicator *pcnaGFP*, and were small on average compared to control CB NBs during earlier

developmental stages (L3 control, average diameter 10-15µm) (**Fig. 1B,C,E**). However, at 30 hours APF when control CB NBs are still present, *N RNAi* CB NBs were larger on average (**Fig. 1B,C,E**). To confirm the *N RNAi* phenotype, we used MARCM to generate CB NB clones mutant for the Notch loss of function allele, *Notch^{55e11}* (Lee and Luo, 1999; Lehmann et al., 1983). Animals were heat shocked at freshly hatched larval stages and brains assayed at 48 hours APF. More than 50% of the GFP positive clones had a single Dpn positive NB (**Fig. 1F**). In contrast, control clones had no Dpn positive NBs. We conclude that Notch functions in a lineage-dependent manner to eliminate CB NBs and terminate neurogenesis.

Next, we assayed other Notch pathway components. Following knockdown of *kuz*, (*worGAL4,UAS-kuzRNAi #HMS05424*), CB NBs, other than the MB NBs, remained at 48 hours APF and in adults (**Fig. 1G,I,J**). Following *Su(H)* knockdown (*worGAL4,UAS-Su(H)RNAi #HMS05748*), CB NBs, other than the MB NBs, also remained (**Fig. 1H,J**). Similar to the *N RNAi* phenotype, ectopically persisting CB NBs expressed Dpn and *pcnaGFP*, and were small (**Fig. 1K**). We conclude that the evolutionarily conserved Notch cell signaling pathway is required for CB NB elimination and neurogenesis termination.

Delta expressed in neighboring GMCs and cortex glia regulates Notch activity in CB NBs

Next, we assayed the expression of Delta (Dl) and Serrate (Ser), two Notch ligands that activate Notch signaling when expressed on neighboring cells. Using a *Delta-GFP* protein trap line, we found that Delta was expressed in CB NBs and their recently born Prospero (Pros) positive GMC progeny during larval stages, consistent with previous reports (**Fig. 2A,B**) (Kooh et al., 1993; Sood et al., 2022). Delta was also expressed in cortex glia, a glial subset that ensheathes CB NBs and their GMC progeny, but levels were relatively low (**Fig. 2C**) (Hayashi et al., 2002; Yuan et al., 2020). Using a *Serrate-GFP* protein trap line, we found that Serrate was expressed in cortex glia, but not in CB NBs nor their GMC progeny (**Fig. 2D-F**). Next, we knocked down each of the Notch ligands to determine which Notch ligand from what cell type regulates Notch activity in CB NBs. We used the NB specific *E(spl)my-GFP* reporter to assay Notch activity (Furriols and Bray, 2001; Almeida and Bray, 2005; Zacharioudaki et al., 2012). In controls, CB NBs express *E(spl)my-GFP* and following knock down of Notch, *E(spl)my-GFP* was not expressed (**Fig. 2G,H**). Next, we used *worGAL4* to knock down Delta in CB NBs and because GAL4 is inherited after CB NBs divide, neighboring GMC progeny as well. Following knock down of Delta (*worGAL4,UAS-Dl RNAi #HMS01309*), *E(spl)my-GFP* was not detected in CB NBs (**Fig. 2I**). Next, we used a pan-glial GAL4 line to knock down either Delta or Serrate in cortex glia. Following knock down of Delta (*repoGAL4,UAS-Dl RNAi #HMS01309*), *E(spl)my-GFP* was reduced compared to controls (**Fig. 2J,K**). In contrast, following Serrate knock down (*repoGAL4,UAS-SerRNAi #HMS01179*), *E(spl)my-GFP* was not affected (**Fig. 2L**). We conclude that Delta is the primary Notch ligand expressed in CB NBs and their GMC progeny, while both Delta and Serrate are expressed in neighboring cortex glia. Moreover, Delta regulates CB NB Notch activity.

Delta-dependent Notch activation is required for CB NB elimination and termination of neurogenesis

Next, we knocked down each of the Notch ligands in neighboring cell types and assayed CB NB number during pupal stages. When Delta was knocked down in NBs and GMC progeny (*worGAL4,UAS-Dl RNAi #HMS01309*), ectopically persisting CB NBs were found in brains at all stages examined and in young adults (**Fig. 3A-C**). When Delta was knocked down in cortex glia (*NP0577GAL4,UAS-Dl RNAi #HMS01309*), one, occasionally two ectopically persisting CB NBs were found at mid pupal stages (**Fig. 3E,F**). Moreover, CB NBs that ectopically persisted tended to be larger than control CB NBs at early stages, consistent with the notion that cell size correlates with timing of termination (**Fig. 3D,G**) (Maurange et al., 2008; Siegrist et al., 2010; Homem et al.,

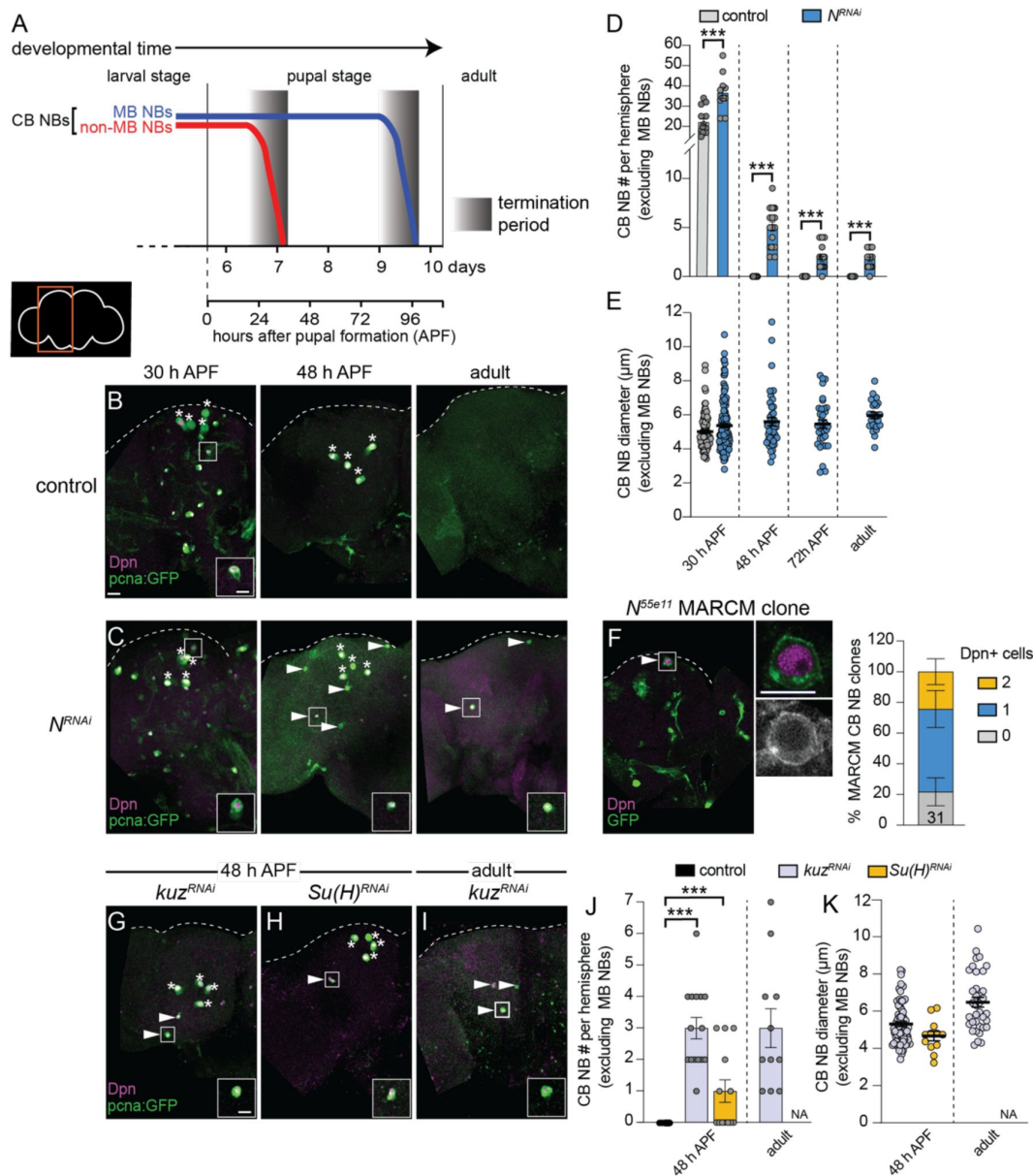


Figure 1

Notch signaling regulates timing of CB NB elimination and neurogenesis termination.

(A) Schematic showing developmental timeframe of CB NB elimination with timeline below used for developmental staging. (B-C, G-I) Maximum intensity projections of single brain hemispheres from indicated genotypes, times, with markers listed in bottom left. Asterisks indicate the four MB NBs and arrowheads indicate some of the ectopically proliferating CB NBs (non-MB NBs). One ectopic CB NB (white box) shown at higher magnification in bottom right. (D,J) Quantification of CB NB number (excluding MB NBs) per brain hemisphere at indicated times and genotypes. Each data point represents one brain hemisphere, mean ± SEM, ***p-values ≤ 0.001 (unpaired two-tailed Student's *t*-test). (E,K) Quantification of average CB NB diameter, used as a proxy for NB size, at indicated times and genotypes. Each data point equals one CB NB (n ≥ 4 animals per genotype), mean ± SEM. (F) Single optical section of a brain hemisphere from indicated genotype at 48 hours APF with markers listed in panels with high magnification panel to right of ectopic CB NB in white box. Distribution of *Notch^{55e11}* MARCM CB NB clones containing Dpn positive NBs. Scale bar equals 20 μm (panels) or 10 μm (insets) in this and all subsequent figures. Panel genotypes listed in **Supplemental Table 1**.

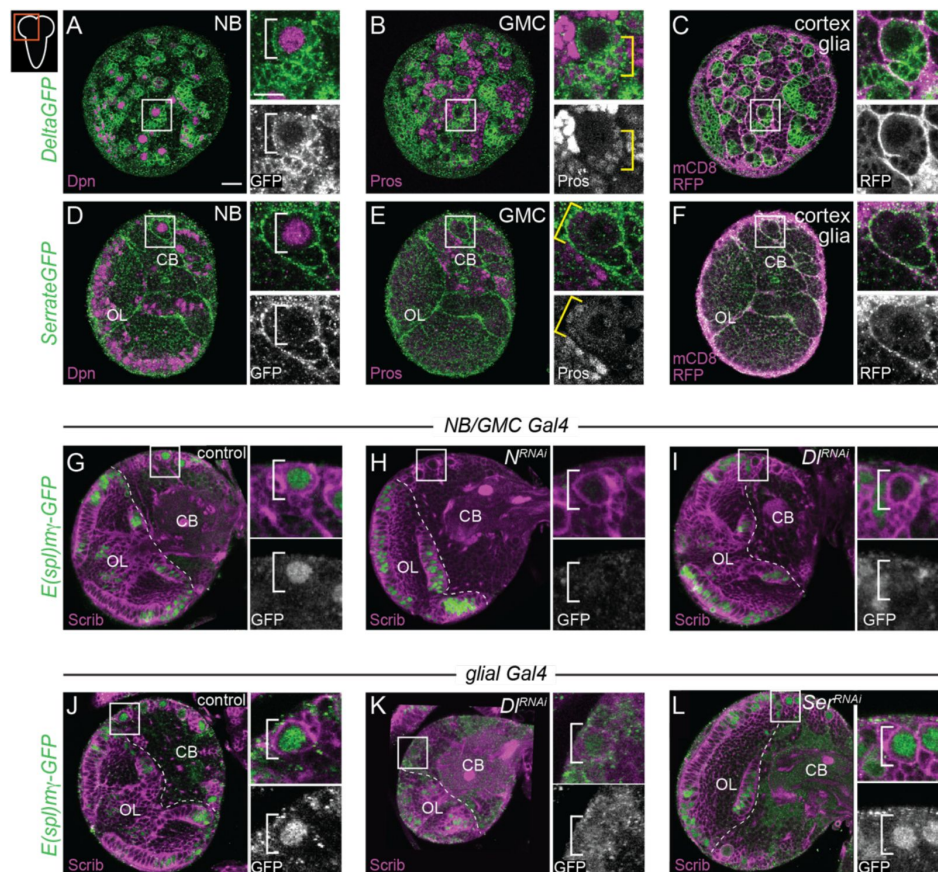


Figure 2

Delta expressed in CB NBs, GMCs, and cortex glia regulates CB NB Notch activity.

(A-F) Single optical section of a brain hemisphere from the indicated genotypes at wandering L3 stages. Higher magnification image of the CB NB highlighted by the white box is shown to the right of the colored overlays. Top panels are higher magnification colored overlay with single channel grayscale images below. White brackets indicate the CB NB and yellow brackets indicate newborn GMC progeny. (G-L) Single optical section of a brain hemisphere from the indicated genotypes at 72 hours ALH. Higher magnification image of the CB NB highlighted by the white box is shown to the right of the colored overlays. Scale bar equals 20 μm (panels) and 10 μm (insets). CB: central brain; OL: optic lobe. Panel genotypes listed in Supplemental Table 1.

2014 [\[link\]](#); Yang et al., 2017 [\[link\]](#)). Next, although Serrate was not detected in CB NBs nor their GMC progeny, we still assayed CB NB number and size after Serrate knock down in both NBs and their GMC progeny (*worGAL4,UAS-SerRNAi #HMS01179*). No differences were observed compared to controls (**Fig. 3H,I** [\[link\]](#)). Next, we knocked down Serrate in glia (*repoGAL4,UAS-SerRNAi #HMS01179*) and again no differences were found compared to controls (**Fig. 3J** [\[link\]](#)). We conclude that Delta, but not Serrate, is required for CB NB elimination and termination of neurogenesis.

Early Notch signaling is required to terminate neurogenesis during pupal stages

Next, we used a temperature sensitive GAL80 to determine when during development Notch signaling is required to eliminate CB NBs and terminate neurogenesis. Animals were raised at 29°C (GAL80 inactive, GAL4 active) until 72 hours ALH and then switched to 18°C (GAL80 active, GAL4 inactive) or the converse (**Fig. 4A,B** [\[link\]](#)). In control animals at 48 hours APF, after either temperature shift regime, only the four MB NBs were present in each brain hemisphere (**Fig. 4C** [\[link\]](#) and data not shown). When Notch or Delta knock down animals were raised at 29°C (Notch pathway inactive) and then switched to 18°C late (Notch pathway active), a significant number of persisting CB NBs were found at 48 hours APF (**Fig. 4A,C** [\[link\]](#)). In contrast, when animals were raised at 18°C (Notch pathway active) and then switched to 29°C late (Notch pathway inactive), no or significantly fewer persisting CB NBs were found (**Fig. 4B,C** [\[link\]](#)). Absence or presence of Notch pathway activity under each temperature shift regime was verified using *E(spl)my-GFP* reporter expression (**Fig. S1A** [\[link\]](#), **B** [\[link\]](#)). We conclude that early Notch pathway activity is required to eliminate CB NBs and terminate neurogenesis.

Delta-dependent Notch activation refines temporal boundaries by closing the early CB NB temporal window

During development, CB NBs sequentially express a series of intrinsic factors over time to generate a diversity of neuron types (**Fig. 5A** [\[link\]](#)). Defects in intrinsic temporal factor expression lead to changes in the molecular composition of neuron types produced and defects in timing of CB NB elimination and neurogenesis termination (Isshiki et al., 2001 [\[link\]](#); Maurange et al., 2008 [\[link\]](#); Liu et al., 2015 [\[link\]](#); Ren et al., 2017 [\[link\]](#); Syed et al., 2017 [\[link\]](#); Yang et al., 2017 [\[link\]](#); Pahl et al., 2019 [\[link\]](#)). Because Notch signaling is required early to eliminate CB NBs, we assayed expression of early temporal factors, Cas and Svp. In freshly hatched control larvae (0 hours ALH), approximately 50% of CB NBs expressed Cas and 5% expressed Svp (**Fig. 5B** [\[link\]](#)). Over time, the percentage of Cas expressing CB NBs declined, while Svp expressing CB NBs modestly increased (**Fig. 5B** [\[link\]](#)). Less than 1% of CB NBs co-expressed Cas and Svp at any stage and expression of both factors was absent by 48 hours ALH (**Fig. 5B,C** [\[link\]](#)). This is consistent with work published previously (Isshiki et al., 2001 [\[link\]](#); Tsuji et al., 2008 [\[link\]](#); Chai et al., 2013 [\[link\]](#); Maurange et al., 2008 [\[link\]](#); Ren et al., 2017 [\[link\]](#); Syed et al., 2017 [\[link\]](#)). When Notch or Delta were knocked down in CB NBs and their GMC progeny, approximately 50% of CB NBs expressed Cas at freshly hatched larval stages (0 hours ALH), same as controls, and slightly more expressed Svp (**Fig. 5B** [\[link\]](#)). Over time, the percentage of Cas expressing CB NBs declined, while Svp expressing CB NBs remained relatively unchanged or reduced (**Fig. 5B** [\[link\]](#)). At 48 hours ALH, in contrast to controls, Cas and Svp were still expressed in CB NBs in both Notch and Delta knockdown animals (**Fig. 5B,D** [\[link\]](#), **E**). This suggests that Notch signaling is required for the cessation of early temporal factor expression. Moreover, early temporal defects could lead to defects in later temporal factor expression.

Next, we looked at later developmental time points. In control animals at 72 hours ALH, some CB NBs still expressed Imp, but most had transitioned to expressing Syp in response to steroid hormone signaling, and by 96 hours ALH, all expressed Syp and E93 (except MB NBs) (**Fig. 5F** [\[link\]](#), **I** [\[link\]](#), **J** [\[link\]](#) and **Fig. S2A** [\[link\]](#), **D** [\[link\]](#)). This is consistent with work published previously (Liu et al., 2015 [\[link\]](#); Ren et al., 2017 [\[link\]](#); Syed et al., 2017 [\[link\]](#); Yang et al., 2017 [\[link\]](#)). When Notch or Delta were

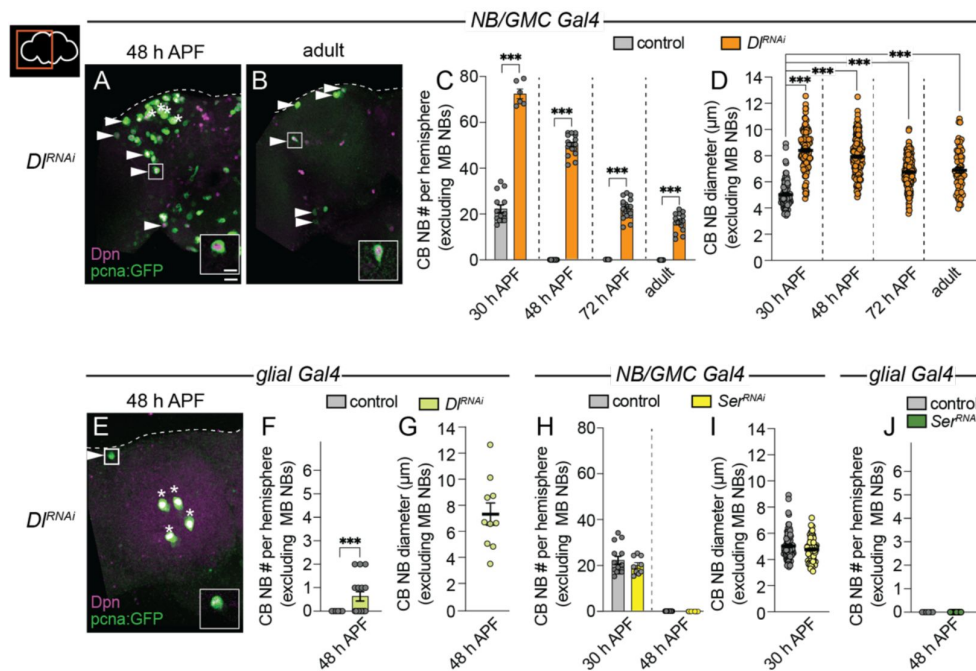


Figure 3

Delta is required to eliminate CB NBs and terminate neurogenesis.

(A,B,E) Maximum intensity projections of single brain hemispheres from indicated genotypes. Asterisks indicate the MB NBs and the white arrowheads indicate some of the ectopically proliferating CB NBs. Inset shows a higher magnification of the ectopically proliferating CB NB highlighted by the white box. (C,F,H,J) Quantification CB NB number (excluding the MB NBs). Each data point represents one brain hemisphere. Control data in (C) is the same as **Figure 1D**. Mean±SEM. *** $P \leq 0.001$ (unpaired two-tailed Student's t -test). (D,G,I) Quantification of CB NB size (excluding the MB NBs) in the indicated genotypes and developmental times. Each data point represents one neuroblast ($n \geq 4$ animals per genotype). Control data in (D) is the same as **Figure 1E**. Mean±SEM. *** $P \leq 0.001$, * $P \leq 0.033$ (Kruskal-Wallis test). Scale bar equals 20 μm (panels) and 10 μm (insets). Panel genotypes listed in **Supplemental Table 1**.

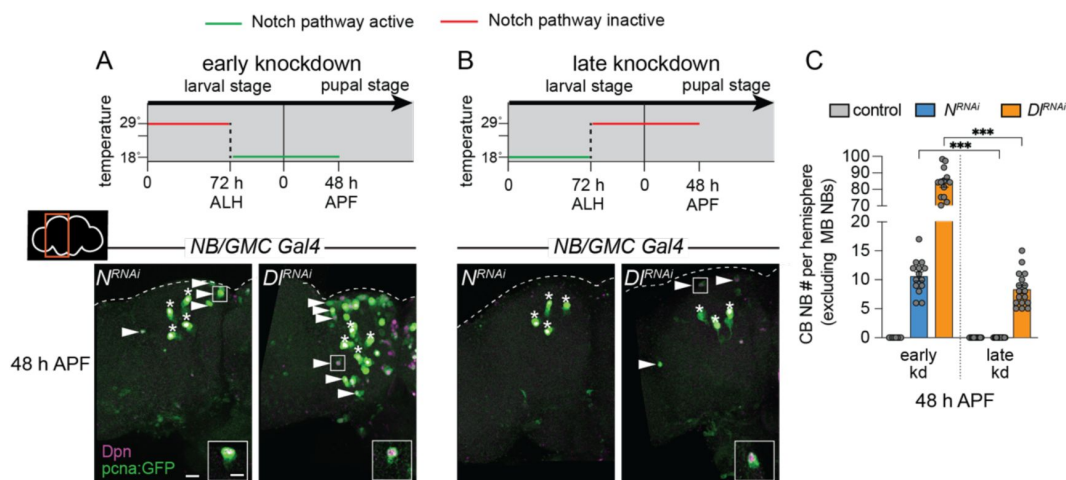


Figure 4

Notch is required early to eliminate CB NBs.

(A,B) Top, schematic depicting experimental setup to temporally control knockdown of Notch signaling. Bottom, maximum intensity projections of single brain hemispheres from indicated genotypes at 48 hours APF relative to 25°C. Asterisks indicate the MB NBs and the white arrowheads indicate some of the ectopically proliferating CB NBs. Inset shows a higher magnification of an ectopically proliferating CB NB highlighted by the white box. (C) Quantification of CB NB number (excluding the MB NBs) in the indicated genotypes. Each data point represents one brain hemisphere. Mean±SEM. **** $P \leq 0.001$ (Welch's t test). Scale bar equals 20 μ m (panels) and 10 μ m (insets). Panel genotypes listed in **Supplemental Table 1**.

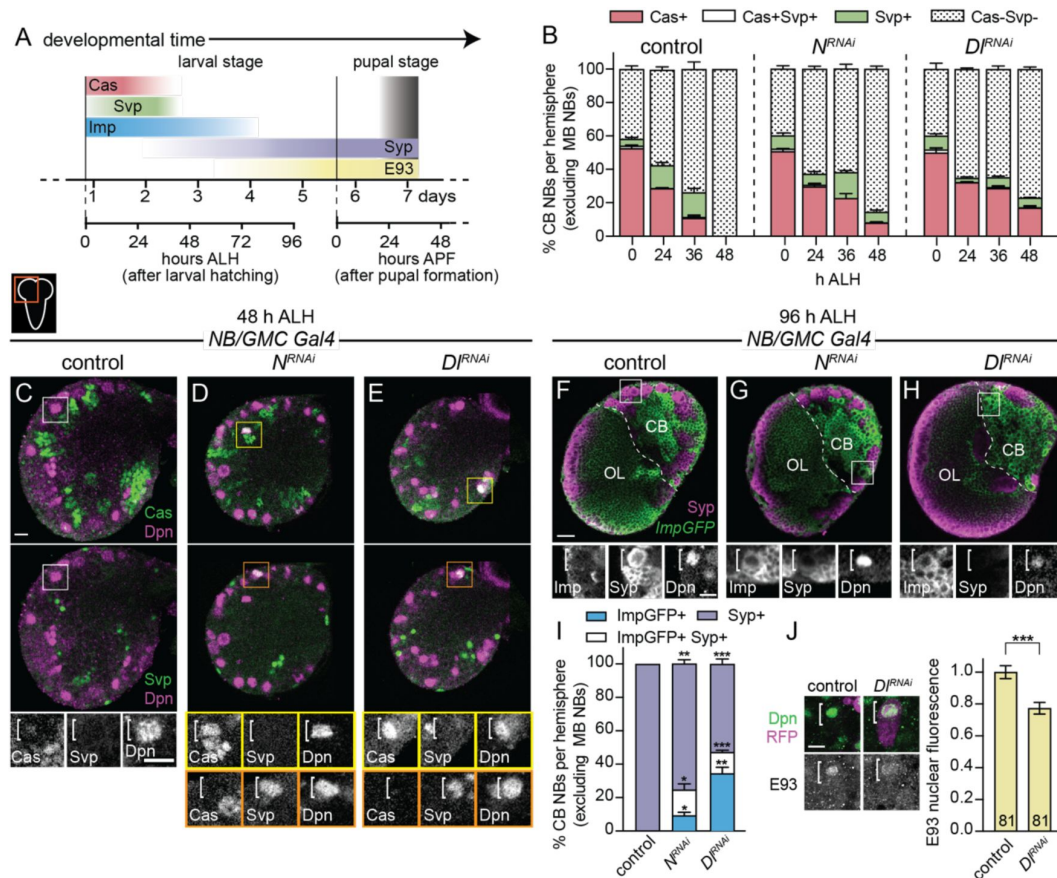


Figure 5

Notch signaling refines temporal factor expression boundaries.

(A) Schematic of temporal factor expression in CB NBs during larval and pupal development. Top timeline (days) refers to developmental timing with two timelines below used for developmental staging. Larva hatch 22 hours after egg lay. (B,I) Quantification of the percentage of CB NBs (excluding the MB NBs) in the indicated genotypes and developmental times expressing the indicated temporal factors. Mean±SEM. $n \geq 3$ animals, *** $P \leq 0.001$, ** $P \leq 0.002$, * $P \leq 0.033$ (Two-way ANOVA). (C-H) Single optical section of a brain hemisphere from the indicated genotypes and developmental times. Higher magnification image of the CB NB highlighted by the white box is shown below the colored overlays. White brackets indicate the CB NB. (J) Single optical section of a CB NB, colored overlay with grayscale image below. White brackets indicate the CB NB with quantification of normalized nuclear E93 intensities. Column numbers indicate the number of CB NB clones (excluding the MB NBs) scored. Mean±SEM. *** $P \leq 0.001$ (Mann-Whitney test). Scale bar equals 20 μ m (panels) and 10 μ m in single CB NB panels. CB: central brain; OL: optic lobe. Panel genotypes listed in [Supplemental Table 1](#).

knocked down in NBs and their GMC progeny, we found that CB NBs still expressed Imp at both 72 and 96 hours ALH (**Fig. 5G-I** and **Fig. S2B-C-D**). Coincident with prolonged Imp expression, we found a reduction in CB NBs expressing Syp and many co-expressed both Imp and Syp, a phenotype not seen in control animals (**Fig. 5G-I** and **Fig. S2B-C-D**). Next, we assayed E93, whose expression is dependent on ecdysone signaling and in the MB NB lineage, Syp (Syed et al., 2017; Pahl et al., 2019). We generated RFP expressing CB NB clones that co-express *Dl RNAi* and found that E93 protein levels were reduced more than 20% compared to controls (**Fig. 5J**). We conclude that Delta-dependent Notch activation is required to sharpen the boundaries of temporal factor expression. Moreover, these defects in temporal factor boundaries could lead to defects in CB NB elimination.

Defects in timing of temporal transitions could be due to defects in cell cycle progression, although embryonic NBs still transition independent of cell division (Grosskortenhaus et al., 2005). We used PH3 to assay CB NB mitotic activity. In Delta knock down animals, the percentage of PH3 positive CB NBs was reduced compared to control (**Fig. S2E**). At 48 h APF however, Delta knock down CB NBs were still dividing based on PH3 expression (**Fig. S2F**). To determine whether CB NBs ectopically persist due to defects in cell cycle rate, we co-expressed *dp110* to constitutively activate PI3-kinase in Delta knock down animals. A significant number of *pnaGFP* expressing, Dpn positive CB NBs were still observed, suggesting that defects in cell cycle timing and growth rates alone cannot account for ectopic persistence of CB NBs into later developmental stages and adulthood (**Fig. S2G**).

CB NBs with reduced Notch pathway activity persist into adulthood due to temporal patterning defects

Next, we assayed temporal factor expression in CB NBs that ectopically persisted into late pupal stages. We generated *Notch^{55e11}* MARCM CB NB clones as described previously and found that ectopically persisting CB NBs expressed either Imp alone, co-expressed both Imp and Syp, or expressed Syp alone similar to CB NBs at earlier larval stages (**Fig. 6A**, *n* = 14 clones). Similar expression profiles were observed in ectopically persisting CB NBs in Delta knock down animals (*worGAL4, UAS-Dl RNAi #HMS01309*) (**Fig. 6B**). Next, we tested whether temporal patterning defects account for the ectopic persistence of CB NBs with reduced Notch pathway activity. First, we knocked down Imp. Knocking down Imp alone leads to premature CB NB loss due to premature expression of late temporal factors (**Fig. S3A-B**) (Yang et al., 2017). When Imp was knocked down together with Delta (*worGAL4, UAS-Dl RNAi #HMS01309, UAS-Imp RNAi #HMS01168*), CB NB number was significantly reduced compared to Delta knockdown alone (**Fig. 6C,D,G**). CB NB size was also reduced, consistent with previous work demonstrating the importance of the early factor Imp in promoting growth (**Fig. 6H**) (Yang et al., 2017). Next, we constitutively expressed the late temporal factor E93, since E93 levels were reduced in *Dl RNAi* CB NB clones. When E93 was constitutively expressed in Delta knock down animals, a significant reduction in CB NB number was observed (**Fig. 6E-H**). We conclude that ectopically persisting CB NBs in animals with reduced Notch pathway activity is due to defects in CB NB temporal patterning: Imp expression is prolonged and E93 levels are reduced.

The early temporal factor Imp positively regulates Delta expression

To better understand how Notch signaling controls CB NB temporal factor expression, we mined publicly available datasets. The datasets that we mined include (1) results from Notch genetic, molecular, and biochemical interaction studies (<https://flybase.org/reports/FBgn0004647>) (2) results from RIP-seq (RNA immunoprecipitation) experiments using Imp or Syp as bait (McDermott et al., 2014; Samuels et al., 2020), and (3) RNA-sequence data from isolated, pooled AL (antennal lobe) NBs at early and late timepoints (Liu et al., 2015). First, we identified

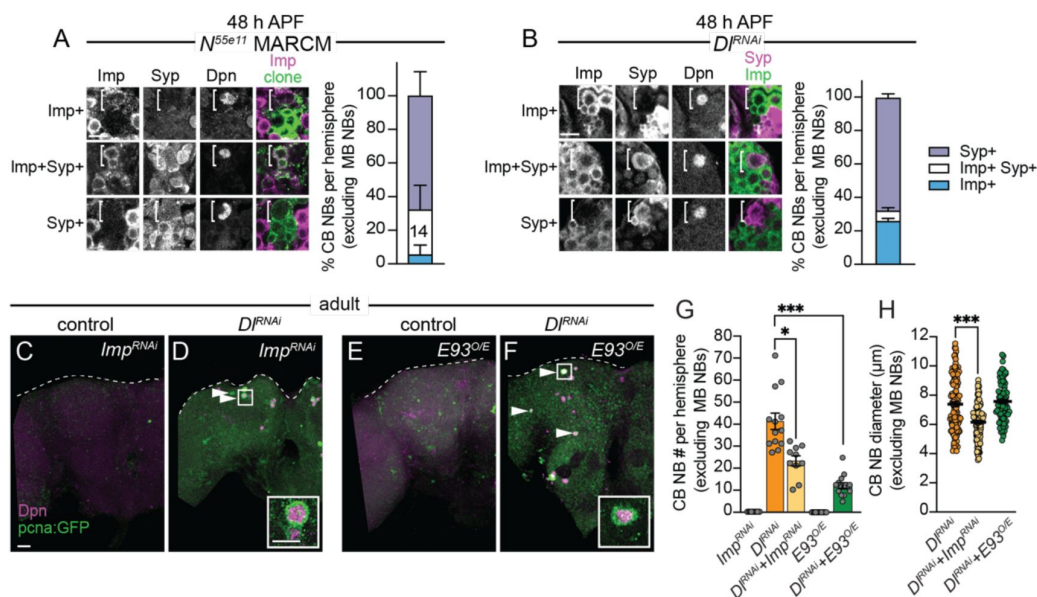


Figure 6

CB NBs ectopically persist due to prolonged early factor Imp expression and reduced late factor E93 expression.

(A,B) Single optical sections of CB NBs from indicated genotypes and developmental times. Single channel grayscale images with colored overlay to the right. White brackets indicate CB NBs. Right, percentage of CB NBs (excluding the MB NBs) in the indicated genotypes and developmental times expressing the indicated temporal factors. Column number (A) indicates number of clones scored. Mean $\pm$ SEM. $n \geq 3$ animals (B). (C-F) Maximum intensity projections of single brain hemispheres from indicated genotypes in one-day old adults. White arrowheads indicate some ectopically proliferating CB NBs (MB NBs are absent). Inset shows a higher magnification of an ectopically proliferating CB NB highlighted by the white box. (G) Quantification of CB NB number (excluding MB NBs) in the indicated genotypes. Each data point represents one brain hemisphere. Mean $\pm$ SEM. *** $P \leq 0.001$, * $P \leq 0.033$ (Kruskal-Wallis ANOVA). (H) Quantification of CB NB size (excluding the MB NBs) in the indicated genotypes. Each data point represents one CB NB ($n \geq 4$ animals per genotype). Mean $\pm$ SEM. *** $P \leq 0.001$, * $P \leq 0.033$ (Kruskal-Wallis ANOVA). Scale bar equals 20 μ m (panels) and 10 μ m (insets). Panel genotypes listed in [Supplemental Table 1](#).

genes that were differentially expressed (DEGs) in AL NBs early (24 hours ALH) versus late (84 hours ALH). We identified 1861 genes (adj. p-value<0.1 and log fold change>1), including known temporal factors (**Fig. 7A**). Next, we asked which if any were present in datasets from Notch genetic, molecular, and biochemical interaction studies. We identified 59 genes (**Fig. 7B,C**). Next, we determined that 23 of these 59 genes were found in the Imp-RIP dataset (**Fig. 7B**, orange highlight) and 6 in the Syp-RIP dataset (**Fig. 7B**, purple highlight). To our surprise, Delta was one of the genes on this gene list. Delta transcript levels were expressed high early, and Delta mRNA co-immunoprecipitated with Imp in RIP-seq experiments. This raised the possibility that Delta, a known Notch target gene, is regulated by NB temporal factors (Zhang et al., 2021). To test this possibility, we knocked down Imp in CB NBs (*worGAL4,UAS-Imp-RNAi #HMS01168*) and assayed expression of the protein trap, *Delta-GFP*. At 24 hours ALH, *Delta-GFP* levels were reduced compared to controls (**Fig. 7D,E**). This suggests that Imp positively regulates Delta.

Next, we asked whether late temporal factors would also regulate Delta, since Delta transcript levels decrease over time. We assayed *Delta-GFP* in CB NBs following constitutive Syp or E93 expression. We examined brains at 24 hours ALH, a time when Syp and E93 are normally not expressed and found reduced *Delta-GFP* expression in NBs and their GMC progeny compared to controls under both conditions (**Fig. 7F,G**). Together with the reduction in Delta transcript levels during late larval stages (**Fig. 7B**), these results suggest that Delta and Delta-dependent Notch transactivation are regulated by CB NB intrinsic temporal factors.

Discussion

Here we report that CB NBs utilize Notch signaling to progress forward through their stem cell lineages, ultimately terminating their divisions through differentiation or death (see Model **Fig. 7H**). Somewhat paradoxically, we find that early Notch activity is required early to terminate CB NB divisions late. This is because Notch regulates early temporal patterning and defects in early temporal patterning transmit to late temporal defects including the time at which CB NBs stop divisions. Notch curtails expression of at least three early temporal factors (Cas, Svp, and Imp), suggesting that Notch may function broadly to close the early temporal window. It is known that the early to late temporal transition is dependent on both early intrinsic temporal factors, Cas and Svp, and on extrinsic steroid hormone signaling (ecdysone) (Ren et al., 2017; Syed et al., 2017). Cas overexpression is sufficient to prolong Svp expression, but Cas is not required for Svp expression, and Svp primes CB NBs to respond to ecdysone (Ren et al., 2017). Whether Notch pathway activity curtails both Cas and Svp or just Cas remains an open question, however it has been reported that both *cas* and *svp* are associated with at least one enhancer that is responsive to Notch activity (Zacharioudaki et al., 2016). Also, it remains unknown whether Notch directly inhibits Imp or whether Notch indirectly inhibits Imp through Syp expression in response to ecdysone or a yet unidentified factor. Notch was recently shown to regulate timing of Sloppy-paired expression in the optic lobe (Ray et al., 2022).

While some CB NBs maintained early Imp expression, others co-expressed both Imp and Syp, or expressed Syp alone. This suggests that Notch function is lineage-dependent and/or suggests that more than one pathway regulates lineage progression. While Cas is likely expressed in all CB NBs, Svp appears to be more restricted. Whether Notch inhibits early temporal progression only in Svp expressing CB NBs is not yet known. Somewhat unexpectedly, we also found a significant percentage of ectopically persisting CB NBs expressing the late temporal factor Syp. Syp promotes accumulation of nuclear Pros in CB NBs during pupal stages to induce terminal differentiation (Maurange et al., 2008; Yang et al., 2017). This suggests that either a Pros-independent mechanism exists to eliminate CB NBs and/or that Syp regulates expression of additional unknown temporal factors required for Pros nuclear accumulation. As we report here, Syp and E93 inhibit expression and localization of Delta and decreased Delta leads to ectopic persistence of CB NBs. This is consistent with the notion that once CB NBs transition from early to late temporal factor

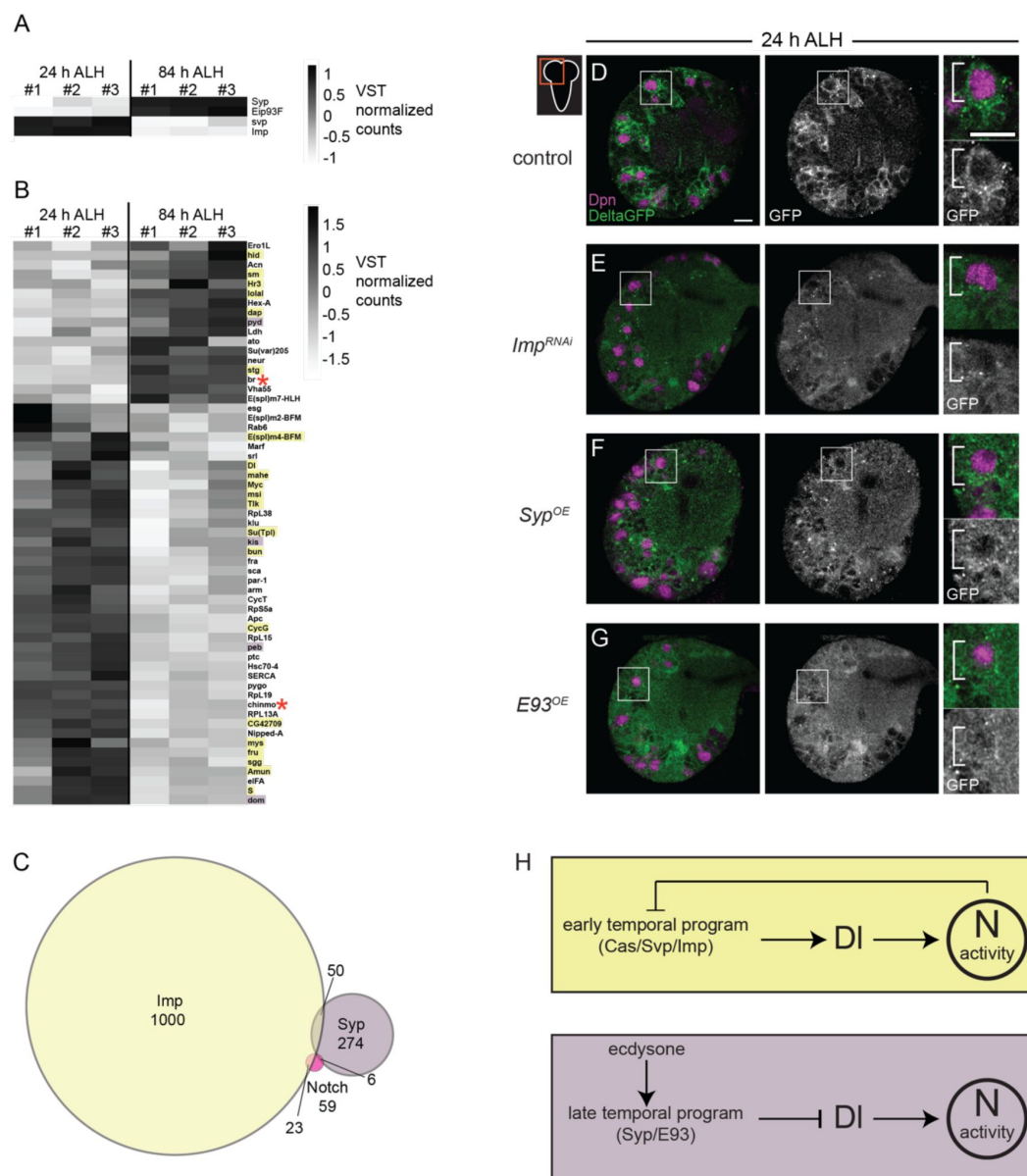


Figure 7

Delta is expressed at higher levels early and is positively regulated by the early temporal factor Imp.

(A-B) Heat map showing the list of genes that are differentially expressed in the AL NBs from 24 to 84 hours ALH. In (B) genes are color coded to show if they are also targets of either Imp (yellow) or Syp (purple) or both (red asterisk). Variance-stabilizing transformation (VST) normalized counts were used to plot the heat maps. (C) Venn diagram showing the number of target genes analyzed that are common between Notch, Imp and Syp. (D-G) Single optical section of a brain hemisphere from the indicated genotypes at 24 hours ALH expressing *Delta-GFP*. Higher magnification image of the CB NB highlighted by the white box is shown to the right of the single channel grayscale images. Top panels are higher magnification colored overlay with single channel grayscale images below. White brackets indicate the NB. Scale bar equals 10 μ m. Panel genotypes listed in [Supplemental Table 1](#). (H) Model of Delta-dependent Notch activation in regulation of CB NB temporal patterning.

expression in response to ecdysone, late temporal factors (Syp/E93) inhibit Delta. Whether this changes Notch activity and/or transcription of Notch target genes is not yet known. Nevertheless, it will be important to identify the Notch transcriptional target genes that regulate lineage progression. One good place to start will be to follow up on the eight transcription factors we identified from data mining.

During early larval stages, CB NBs reactivate from quiescence and produce GMCs that express Delta (Zacharioudaki et al., 2012 [DOI](#); Sood et al., 2022 [DOI](#)). This leads to the transactivation of Notch in CB NBs (Sood et al., 2022 [DOI](#)). The early temporal factor Imp positively regulates Delta in CB NBs and their GMC progeny. In developing egg chambers, Imp positively regulates Notch pathway activity by controlling Kuz localization (Fic et al., 2019 [DOI](#)). Whether Imp in CB NBs also regulates Kuz localization remains an open question. Delta is also expressed in cortex glia and regulates CB NB Notch activity. Whether Delta expression in cortex glia changes over time as is the case for CB NBs remains an open question. Hedgehog signaling in CB NBs promotes lineage progression downstream of Cas and Hedgehog ligands are produced in cortex glia and GMCs (Chai et al., 2013 [DOI](#)). Thus, CB NBs integrate cues from their GMC progeny and neighboring cortex glial cells to control temporal progression and lineage termination.

Although the exact course of temporal progression is yet to be defined in the mammalian nervous system, mammalian NSCs temporally express several factors including microRNAs, mRNA-binding proteins and transcription factors, allowing them to produce deep layer neurons (early-born neural progeny), superficial layer neurons (late-born neural progeny) and glial cells sequentially throughout development (Okano and Temple, 2009 [DOI](#); MuhChyi et al., 2013 [DOI](#); Oberst et al., 2019 [DOI](#); Telley et al., 2019 [DOI](#)). COUP-TFI and COUP-TFII, orthologs of *Drosophila* Svp, function as late temporal factors allowing NSCs to switch from producing early-born neural fates to late-born neural fates (Naka et al., 2008 [DOI](#)). This is similar to the function of Svp in the *Drosophila* brain where Svp mutants failed to switch from early Chinmo positive daughters to late Broad-complex positive daughters (Maurange et al., 2008 [DOI](#)). Similarly, mammalian NSCs temporally express RNA-binding protein Imp-1, ortholog of *Drosophila* Imp, and in Imp-1 deficient animals, NSCs are lost prematurely similar to premature loss of CB NBs seen in *Drosophila* (Nishino et al., 2013 [DOI](#); Yang et al., 2017 [DOI](#); Pahl et al., 2019 [DOI](#)). As in *Drosophila* NBs, temporal progression of mammalian NSCs is not completely dependent on cell-intrinsic cues but also requires cell-extrinsic cues like feedback signals from the lineage and environmental cues (Okamoto et al., 2016 [DOI](#); Oberst et al., 2019 [DOI](#); Zhang et al., 2020 [DOI](#)), however, very little is known about the signaling pathways regulating the transitions from early to late temporal fates. Even though Notch activity is required for the temporal switch from neurogenesis to gliogenesis in mammalian NSCs, it remains unclear whether Notch function extends to regulation of temporal progression required for the switch from early born to late born neuron subtypes (Morrison et al., 2000 [DOI](#); Grandbarbe et al., 2003 [DOI](#); Ohtsuka and Kageyama, 2019 [DOI](#)).

Materials and Methods

Fly Stocks

Fly stocks used in this study and their source are listed in the **Supplemental Table 2** [DOI](#).

Animal husbandry

All animals were raised in uncrowded conditions at 25°C with the exception of animals with tub-GAL80(ts). For experiments using *tub-Gal80^{ts}* (temperature sensitive), animals were kept in uncrowded conditions at 29°C and dissected at developmental timings to be equivalent to development at 25°C unless otherwise stated. Animals were staged from hatching for larval dissections and from white prepupae for pupal dissections.

Induction of clones

For induction of Flp-FRT and MARCM clones, animals were heat shocked at 37°C between 30-60 mins at L1 and dissected at the stated developmental timings.

Temperature-shift experimental paradigm

For early knockdown experiments, animals were raised at 29°C until 72 hours ALH (equivalent to 25°C) and then moved to 18°C to develop until 48 hours APF (equivalent to 25°C). For late knockdown experiments, animals were raised at 18°C until 72 hours ALH (equivalent to 25°C) and then moved to 29°C to develop until 48 hours APF (equivalent to 25°C).

Immunofluorescence and confocal imaging

Larval, pupal, and adult brains were dissected as previously described (Pahl et al., 2019 [DOI](#)). In brief, dissected tissues were fixed in 4% EM grade formaldehyde in PEM buffer for 20 mins. (larvae) or 30 mins. (pupae and adults) and rinsed in 1X PBS with 0.1% Triton X-100 (PBT). Tissues were blocked overnight at 4°C in 10% normal goat serum in PBT followed by antibody staining. Primary antibodies used are listed in the **Supplemental Table 2** [DOI](#). To detect primary antibodies, Alexa-Fluor conjugated secondary antibodies (Thermo-Fisher) listed in the **Supplemental Table 2** [DOI](#) were used. Images encompassing the entire brain hemispheres were acquired using a Leica SP8 laser scanning confocal microscope equipped with a 63x/1.4 NA and 40x/1.3 NA oil-immersion objectives and analyzed using Fiji software. All images were processed using Fiji and Adobe Photoshop software and figures assembled using Adobe Illustrator software. NBs were identified based on Dpn expression and superficial location. The Fiji “cell counter” plugin was used to count and track number of Dpn positive NBs. NB size was calculated by averaging the lengths of two perpendicular lines through the center of the NB in Fiji. Quantification of fluorescence was performed in Fiji. Nuclear E93 levels were quantified as follows: CB NB nuclei labeled with Dpn were manually traced and the average E93 fluorescence intensity measured in the nucleus.

Normalized E93 nuclear fluorescence intensity was determined as a ratio of nuclear E93 fluorescence intensity in a clone NB to nuclear E93 fluorescence intensity in a control NB in the same z-plane. All data is represented as mean $\pm$ standard error of the mean and statistical significance was determined using unpaired two-tailed Student's t-tests or ANOVAs in Prism 9.

RNA-Seq Data Analysis

Differential Gene Expression (DGE) analysis was performed using the publicly available dataset published in Liu et al., 2015 [DOI](#) (GSE71103). We aligned FASTQ data to a reference Dm genome using STAR (version 2.7.2b) (Dobin et al., 2013 [DOI](#)). Read count tables were generated using the htseq-count package from bioconda, and DGE was performed using DESeq2 (Love et al., 2014 [DOI](#)). We made contrasts between 24h ALH and 84h ALH antennal lobe NBs in order to identify differentially expressed genes of interest (adjPvalue <0.1 and log Fold Change >1). The list of 1861 differentially expressed genes was compared to the list of downstream genetic interactions of Notch pathway from FlyBase (<https://flybase.org/reports/FBgn0004647> [DOI](#)) to obtain the list of 59 differentially expressed genes of interest. Finally, VST normalized counts of the 59 genes were used to generate the heatmaps using heatmap.2 from the ggplot2 package (<https://ggplot2.tidyverse.org> [DOI](#)). To obtain common target genes between Notch and Imp, we compared our list of 59 genes to the top 1000 Imp target genes obtained via RIP-seq by Samuels et al., 2020 [DOI](#). We performed similar comparison on the 274 Syp target genes obtained via RIP-seq by McDermott et al., 2014 and our list of 59 genes to obtain common Notch and Syp target genes.

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author contributions: C.S., conception and design of work, acquisition, analysis, and data interpretation, drafting and revision of manuscript. Md A.N., K.R.B., M.C.P., S.E.D. data acquisition. S.E.S. conception and design, data interpretation, drafting and revision of manuscript. This work was supported by NIH grant R35 GM141886.

Figure	Genotype	
Fig. 1B	<i>worGAL4/+ (OregonR) ; pcnaGFP/+ (OregonR)</i>	control
Fig. 1C	<i>worGAL4/+; pcnaGFP/UAS-NotchRNAi</i>	
Fig. 1F	<i>hsflp, tubgal80, FRT19A/Notch55e11 FRT19A ; tubGal4, UAS-mCD8GFP/+</i>	
Fig. 1G,I	<i>worGAL4/+; pcnaGFP/UAS-KuzRNAi</i>	
Fig. 1H	<i>worGAL4/+; pcnaGFP/UAS-Su(H)RNAi</i>	
Fig. 2A,B,C	<i>NP0577GAL4/+; UAS-mCD8RFP/+; DeltaGFP/+</i>	
Fig. 2D,E,F	<i>repoGAL4, UAS-mCD8RFP/SerrateGFP</i>	
Fig. 2G	<i>worGAL4/+; E(spl)mγGFP/+</i>	control
Fig. 2H	<i>worGAL4/+; E(spl)mγGFP/UAS-NotchRNAi</i>	
Fig. 2I	<i>worGAL4/+; E(spl)mγGFP/UAS-DeltaRNAi</i>	
Fig. 2J	<i>repoGAL4/+; E(spl)mγGFP/+</i>	control
Fig. 2K	<i>repoGAL4/+; E(spl)mγGFP/UAS-DeltaRNAi</i>	
Fig. 2L	<i>repoGAL4/+; E(spl)mγGFP/UAS-SerrateRNAi</i>	
Fig. 3A	<i>worGAL4/+; pcnaGFP/UAS-DeltaRNAi</i>	
Fig. 3B	<i>worGAL4/+; pcnaGFP/UAS-DeltaRNAi</i>	
Fig. 3E	<i>NP0577GAL4/+; pcnaGFP/UAS-DeltaRNAi</i>	
Fig. 4A	<i>worGAL4,tubGAL80(ts)/+; pcnaGFP/UAS-NotchRNAi</i> (left panel)	
Fig. 4A	<i>worGAL4,tubGAL80(ts)/+; pcnaGFP/UAS-DeltaRNAi</i> (right panel)	
Fig. 4B	<i>worGAL4,tubGAL80(ts)/+; pcnaGFP/UAS-NotchRNAi</i> (left panel)	
Fig. 4B	<i>worGAL4,tubGAL80(ts)/+; pcnaGFP/UAS-DeltaRNAi</i> (right panel)	
Fig. 5C	<i>worGAL4/+ (OregonR)</i>	control
Fig. 5D	<i>worGAL4/+; UAS-NotchRNAi/+</i>	
Fig. 5E	<i>worGAL4/+; UAS-DeltaRNAi/+</i>	
Fig. 5F	<i>ImpGFP/+ (OregonR); worGAL4/+ (OregonR)</i>	control
Fig. 5G	<i>ImpGFP/+; worGAL4/+; UAS-NotchRNAi/+</i>	
Fig. 5H	<i>ImpGFP/+; worGAL4/+; UAS-DeltaRNAi/+</i>	
Fig. 5J	<i>hsFlp; repoGAL80; UAS-DeltaRNAi/Act5c-FRT-CD2-FRT-Gal4, UAS-RFP</i>	
Fig. 6A	<i>hsflp, tubgal80, FRT19A/Notch55e11 FRT19A ; tubGal4, UAS-mCD8GFP/+</i>	
Fig. 6B	<i>worGAL4/+; UAS-DeltaRNAi/+</i>	
Fig. 6C	<i>worGAL4/+; pcnaGFP/UAS-ImpRNAi</i>	
Fig. 6D	<i>pcnaGFP; worGAL4,tubGAL80(ts)/+; UAS-DeltaRNAi/UAS-ImpRNAi</i>	
Fig. 6E	<i>worGAL4/+; pcnaGFP/UAS-Eip93F WT</i>	
Fig. 6F	<i>pcnaGFP; worGAL4,tubGAL80(ts)/+; UAS-DeltaRNAi/UAS-ImpRNAi</i>	
Fig. 7D	<i>DeltaGFP/+</i>	control
Fig. 7E	<i>worGAL4/+; DeltaGFP/UAS-ImpRNAi</i>	
Fig. 7F	<i>worGAL4/UAS-Syp-RB-HA; DeltaGFP/+</i>	
Fig. 7G	<i>worGAL4/+; DeltaGFP/UAS-Eip93F WT</i>	
Supplementary Figures		

Supplemental Table 1

***Drosophila* genotypes per figure panel**

Fig. S1A,B	<i>worGAL4,tubGAL80(ts)/+; E(spl)mγGFP/UAS-DeltaRNAi</i>	
Fig. S2A	<i>ImpGFP/+ (OregonR); worGAL4/+ (OregonR)</i>	control
Fig. S2B	<i>ImpGFP/+; worGAL4/+; UAS-NotchRNAi/+</i>	
Fig. S2C	<i>ImpGFP/+; worGAL4/+; UAS-DeltaRNAi/+</i>	
Fig. S2E	control: <i>worGAL/+ (Oregon R)</i> <i>worGAL4/+; UAS-DeltaRNAi/+</i>	
Fig. S2F	<i>prosGAL4/+; UAS-DeltaRNAi/+</i>	
Fig. S2G	<i>worGAL4,UAS-dp110/+; UAS-DeltaRNAi/+</i>	
Fig. S3A	<i>worGAL4/+; pcnaGFP/UAS-ImpRNAi</i>	

Supplemental Table 1 (continued)

Reagent	Source	Identifier
ANTIBODIES		
rat ant-Dpn (1:100)	Abcam	ab195173
chicken anti-GFP (1:500)	Abcam	ab13970
rabbit anti-dsRed (1:1000)	Clontech	632496
rabbit anti-PH3 (1:1000)	Millipore	06-570
mouse anti-Repo (1:5)	Developmental Studies Hybridoma Bank	8D12
mouse anti-Prospero (1:1000)	Developmental Studies Hybridoma Bank	MR1A
mouse anti-Svp (1:10)	Developmental Studies Hybridoma Bank	5B11
mouse anti-Dlg (1:40)	Developmental Studies Hybridoma Bank	4F3
rabbit anti-Scribble (1:500)	Gift from Chris Q. Doe	
rabbit anti-Cas (1:500)	Gift from Chris Q. Doe	
rabbit anti-Syp (1:250)	Gift from Chris Q. Doe	
guinea pig anti-E93 (1:250)	Gift from Chris Q. Doe	
rabbit anti-Imp (1:250)	Gift from Paul MacDonald	
rat anti-Imp (1:250)	Gift from Claude Desplan	
guinea pig anti-Dpn (1:1000)	Gift from Claude Desplan	
goat anti-chicken Alexa 488 (1:300)	Thermo Fisher Scientific	A32931
goat anti-rat Alexa 555 (1:300)	Thermo Fisher Scientific	A48263
goat anti-rat Alexa 647 (1:300)	Thermo Fisher Scientific	A48265
goat anti-rabbit Alexa 405 (1:300)	Thermo Fisher Scientific	A48254
goat anti-rabbit Alexa 555 (1:300)	Thermo Fisher Scientific	A21428
goat anti-rabbit Alexa 633 (1:300)	Thermo Fisher Scientific	A21071
goat anti-mouse Alexa 405 (1:300)	Thermo Fisher Scientific	A48255
goat anti-mouse Alexa 488 (1:300)	Thermo Fisher Scientific	A11001
goat anti-mouse Alexa 555 (1:300)	Thermo Fisher Scientific	A32727
goat anti-guinea pig Alexa 488 (1:300)	Thermo Fisher Scientific	A11073
goat anti-guinea pig Alexa 555 (1:300)	Thermo Fisher Scientific	A21435
CHEMICALS		
SlowFade™ Diamond antifade reagent	Invitrogen	Catalog # S36963
SlowFade™ Gold antifade reagent	Invitrogen	Catalog # S36937

Supplemental Table 2

Key Resources

Normal Goat Serum	Thermo Fisher Scientific	Catalog # 31873
Paraformaldehyde 16% solution EM grade	Electron Microscopy Sciences	Catalog # 15710
Schneider's Drosophila media	Gibco	Catalog # 21720-024
Triton X-100	Sigma	Catalog # T9284

SOFTWARE		
ImageJ/Fiji	Fiji	http://fiji.sc/
LAS AF	Leica Microsystems	https://www.leica-microsystems.com/products/microscope-software/details/product/leica-las-x-ls/
Prism 9	Graphpad	https://www.graphpad.com/scientific-software/prism/
Photoshop 2022	Adobe	https://www.adobe.com/products/photoshop.html
Illustrator 2022	Adobe	https://www.adobe.com/products/illustrator.html
R-studio	R-studio	https://www.rstudio.com/

EXPERIMENTAL MODEL: <i>Drosophila melanogaster</i>		
Oregon R	Bloomington Drosophila Stock Center	5
<i>wor-Gal4</i>	(Albertson and Doe, 2003)	
<i>tubulin-Gal80(ts)</i>	Bloomington Drosophila Stock Center	7108
<i>repo-Gal4</i>	Bloomington Drosophila Stock Center	7415
<i>NP0577-Gal4</i>	Kyoto Stock Center	112228
<i>repo-Gal80</i>	(Awasaki et al., 2008)	
<i>UAS-Notch RNAi (HMS00001)</i>	Bloomington Drosophila Stock Center	33611
<i>UAS-Kuzbanian RNAi (HMS05424)</i>	Bloomington Drosophila Stock Center	66958
<i>UAS-Su(H)RNAi (HMS05748)</i>	Bloomington Drosophila Stock Center	67928
<i>UAS-Delta RNAi (HMS01309)</i>	Bloomington Drosophila Stock Center	34322
<i>UAS-Serrate RNAi (HMS01179)</i>	Bloomington Drosophila Stock Center	34700

Supplemental Table 2 (continued)

<i>UAS-dp110</i>	Bloomington Drosophila Stock Center	25914
<i>Delta-GFP</i>	Bloomington Drosophila Stock Center	59819
<i>Serrate-GFP</i>	Bloomington Drosophila Stock Center	59824
<i>pcna-GFP</i>	(Thacker et al., 2003)	
<i>E(spl)mγ-GFP</i>	(Almeida and Bray, 2005)	
<i>Imp-GFP</i>	Bloomington Drosophila Stock Center	60237
<i>UAS-Imp RNAi (HMS01168)</i>	Bloomington Drosophila Stock Center	34977
<i>UAS-Syp-RB-HA</i>	gift from Tzumin Lee	
<i>UAS-Eip93F WT</i>	Zurich FlyORF	F000587
<i>hsFlp (on X)</i>	gift from Iswar Hariharan	
<i>Act5c-FRT-CD2-FRT-Gal4, UAS-RFP</i>	Bloomington Drosophila Stock Center	30558
<i>Act5c-FRT-CD2-FRT-Gal4, UAS-GFP</i>	gift from Iswar Hariharan	
<i>UAS-mCD8-mRFP</i>	Bloomington Drosophila Stock Center	27399
<i>UAS-mCD8-mGFP</i>	Bloomington Drosophila Stock Center	5137
<i>Notch55e11 FRT19A</i>	Bloomington Drosophila Stock Center	28813
<i>hsflp, tubgal80, FRT19A; tubGal4, UASmCD8GFP</i>	gift from Ben Ohlstein	

Supplemental Table 2 (continued)

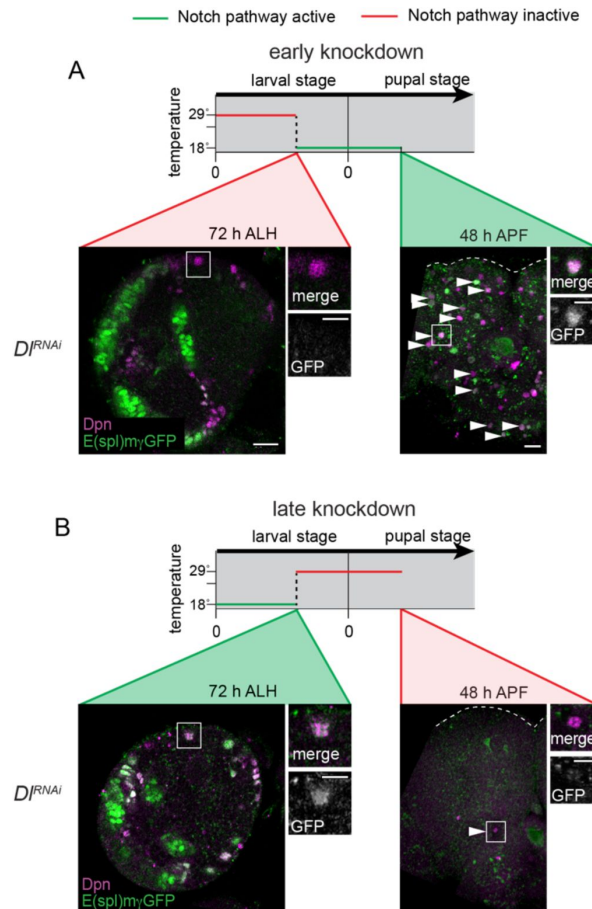


Figure S1

Temporal control of Notch activity in CB NBs. Related to Figure 4.

(A-B) Cartoon above depicts temperature shift regime. Below, single optical sections of brain hemispheres from indicated genotypes and developmental times, with markers listed in panels. Higher magnification image of the CB NB highlighted by the white box is shown to the right of the colored overlays. White arrow heads indicate some of the ectopically persisting CB NBs. Scale bar equals 20 μm (panels) and 10 μm (inset). Panel genotypes listed in **Supplemental Table 1**.

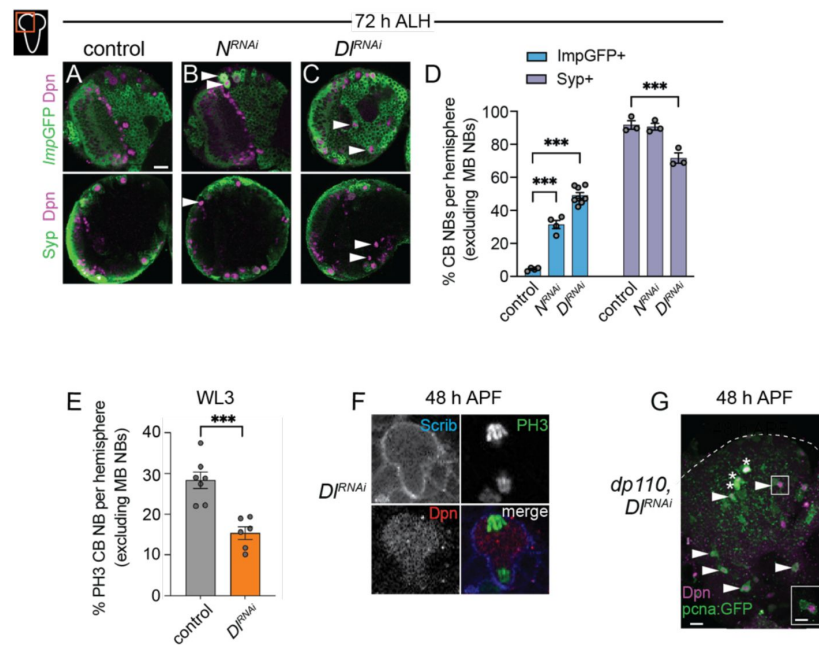


Figure S2

Imp expression is prolonged and Syp expression delayed in animals with reduced Notch pathway activity. Related to Figure 5

(A-C) Single optical section of a brain hemisphere from the indicated genotypes at 72 hours ALH, with markers listed to left. White arrowheads indicate CB NBs with ectopic expression of either Imp-GFP or no Syp expression. (D) Quantification of the percentage of CB NBs (excluding the MB NBs) in the indicated genotypes at 72 hours ALH expressing the indicated temporal factors. Mean $\pm$ SEM. $n \geq 3$ animals, *** $P \leq 0.001$ (Two-way ANOVA). (E) Quantification of the percentage of PH3 positive CB NBs (excluding the MB NBs) in the indicated genotypes at WL3 (wandering L3 stage). (F) Image of a mitotic non-MB NB at 48 hours ALH in DI RNAi expressing animals. (G) Maximum intensity projection of a single brain hemisphere from indicated genotype and time with markers listed in bottom left. Asterisks indicate the MB NBs and arrowheads indicate some of the ectopically proliferating CB NBs (non-MB NBs). One ectopic CB NB (white box) shown at higher magnification in bottom right. Scale bar (A,G) equals 20 μ m and inset (G) equals 10 μ m. Panel genotypes listed in **Supplemental Table 1**.

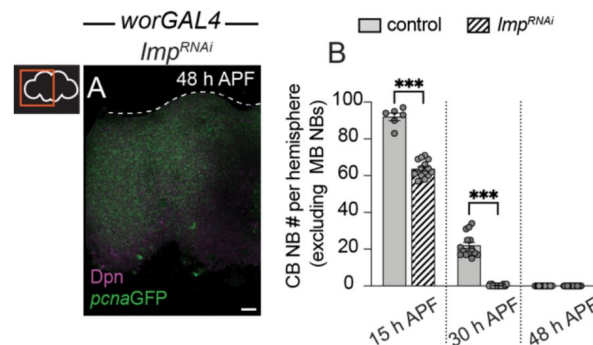


Figure S3

Defects in temporal patterning account for defects in timing of CB NB elimination and neurogenesis termination when Notch activity is reduced. Related to Figure 6

(A) Maximum intensity projection of single brain hemisphere from indicated genotype at indicated time. (B) Quantification of CB NB number (excluding the MB NBs) in the indicated genotypes. Each data point represents one brain hemisphere. Mean $\pm$ SEM. *** $P \leq 0.001$, ** $P \leq 0.002$, (Kruskal-Wallis ANOVA). Scale bar equals 20 μ m. Panel genotypes listed in **Supplemental Table 1**.

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Reviewer #1 (Public Review):

In this manuscript, the authors are building on their previous work showing Delta-Notch regulates the entrance and exit from embryo-larval quiescence of neural stem cells of the central brain (called CB neuroblasts (NB) (PMID: 35112131)). Here they show that continuous depletion of Notch in NBs from early embryogenesis leads to cycling NBs in the adult. This - cycling NBs in the adult - is not seen in controls. The assumption here is that these Notch-RNAi NBs in adults are those that did not undergo terminal differentiation in pupal development. The authors show that Notch is activated by its ligand Delta which is expressed on the GMC daughter cell and on cortex glia. They determine that the temporal requirement for Notch activity is 0-72 hours after larval hatching (ALH) (i.e., 1st instar through mid-3rd instar at 25°C). In NBs/GMCs depleted for Notch, early temporal markers were still expressed at time points when they should be off and late markers were delayed in expression. These effects were observed in ~20-40% of NBs (Figures 5 and 6). Through mining existing data sets, they found that the early temporal factor Imp - an RNA binding protein - can bind Delta mRNA. They show that Delta transcripts decrease over time, leading to the hypothesis that Delta mRNA is repressed by the late temporal factors. Over-expressing late factors Syp or E93 earlier in development leads to downregulation of a Delta::GFP protein trap. These results lead to a model in which Notch regulates expression of early temporal factors and early temporal factors regulate Notch activity through translation of Delta mRNA.

There are several strengths of this study and no major weaknesses. The authors report rigorous measurements and statistical analyses throughout the study. Their conclusions are appropriate for the results. Data mining revealed an important mechanism - that Imp binds Delta mRNA - supporting the model that that early temporal factors promote Delta expression, which in turn promotes Notch signaling.

An appraisal: The authors use temperature shifts with Gal80TS to show that Notch is required between 0-72 hours ALH. They show with the use of known markers of the temporal factors and Delta protein trap, that Imp promotes Delta protein expression and the later temporal factors reduce Delta, although the molecular mechanisms are not clearly delineated. Overall, these data support their model that the reduction of Delta expression during larval development leads to a loss of Notch activity.

As noted in the Discussion, this study raises many questions about what Notch does in larval CB NBs. For example, does it inhibit Castor or Imp? Is Notch required in certain neural lineages and not others. These studies will be of interest in the community of developmental neurobiologists.

Reviewer #2 (Public Review):

As I indicated in the initial review, the experiments are well conceived and executed, and the data are clear. I also agree with the authors that this work represents a key first step toward understanding how Notch signaling contributes to temporal control of fly neuroblasts. It is my opinion that the authors fall short of demonstrating how Notch signaling and temporal identity genes at the chromatin levels. I find this disappointing given the availability of various tools for looking at dynamic regulation of gene activity at high resolution. Given these weaknesses, my opinion is that the study is descriptive and lacks mechanistic explanation.

Reviewer #3 (Public Review):

In this study, the authors investigate the effects of Notch pathway inactivation on the termination of *Drosophila* neuroblasts at the end of development. They find that termination is delayed, while temporal patterning progression is slowed down. Forcing temporal patterning progression in a Notch pathway mutant restores correct timing of neuroblast elimination. Finally they show that Imp, an early temporal patterning factor promotes Delta expression in neuroblast lineages. This indicates that feedback loops between temporal patterning and lineage-intrinsic Notch activity fine tunes timing of early to late temporal transitions and is important to schedule NB termination at the end of development.

The study adds another layer of regulation that finetunes temporal progression in *Drosophila* neural stem cells. This mechanism appears to be mainly lineage intrinsic - Delta being expressed from NBs and their progeny, but also partly niche-mediated - Delta being also expressed in glia but with a minor influence. Together with a recent study (PMID: 36040415), this work suggests that Notch signaling is a key player in promoting temporal progression in various temporal patterning system. As such it is of broad interest for the neuro-developmental community.

Strengths

The data are based on genetic experiments which are clearly described and mostly convincing. The study is interesting, adding another layer of regulation that finetunes temporal progression in *Drosophila* neural stem cells. This mechanism appears to be mainly lineage intrinsic - Delta being expressed from NBs and their progeny, but also partly niche-mediated - Delta being also expressed in glia but with a minor influence. A similar mechanism has been recently described, although in a different temporal patterning system (medulla neuroblasts of the optic lobe - PMID: 36040415). It is overall of broad interest for the neuro-developmental community.

Weaknesses

The mechanisms by which Notch signaling regulates temporal patterning progression are not investigated in details. For example, it is not clear whether Notch signaling directly regulates temporal patterning genes, or whether the phenotypes observed are indirect (for example through the regulation of the cell-cycle speed). The authors could have investigated whether temporal patterning genes are directly regulated by the Notch pathway via ChIP-seq of Su(H) or the identification of potential binding sites for Su(H) in enhancers. A similar approach has been recently undertaken by the lab of Dr Xin Li, to show that Notch signaling regulates sequential expression of temporal patterning factors in optic lobes neuroblasts (PMID: 36040415), which exhibit a different temporal patterning system than central brain neuroblasts in the present study. As such, the mechanistic insights of the study are limited.

Author Response

The following is the authors' response to the original reviews.

We very much appreciate the constructive comments provided by the reviewers. We have incorporated many of their suggestions and believe the manuscript is much improved.

In brief, we updated the text as suggested and have included three additional panels in supplementary fig. S2E-G. This additional data provides further support that the ectopically persisting neuroblasts are actively dividing and that cell cycle defects alone do not account for temporal patterning phenotypes.

Reviewer #1 (Public Review):

In this manuscript, the authors are building on their previous work showing Delta-Notch regulates the entrance and exit from embryo-larval quiescence of neural stem cells of the central brain (called CB neuroblasts (NB) (PMID: 35112131)). Here they show that continuous depletion of Notch in NBs from early embryogenesis leads to cycling NBs in the adult. This - cycling NBs in the adult - is not seen in controls. The assumption here is that these Notch-RNAi NBs in adults are those that did not undergo terminal differentiation in pupal development. The authors show that Notch is activated by its ligand Delta which is expressed on the GMC daughter cell and on cortex glia. They determine that the temporal requirement for Notch activity is 0-72 hours after larval hatching (ALH) (i.e., 1st instar through mid-3rd instar at 25C). In NBs/GMCs depleted for Notch, early temporal markers were still expressed at time points when they should be off and late markers were delayed in expression. These effects were observed in ~20-40% of NBs (Figures 5 and 6). Through mining existing data sets, they found that the early temporal factor Imp - an RNA binding protein - can bind Delta mRNA. They state that Delta transcripts decrease over time (without any reference to a Figure or to published work), leading to the hypothesis that Delta mRNA is repressed by the late temporal factors. Over-expressing late factors Syp or E93 earlier in development leads to downregulation of a Delta::GFP protein trap. These results lead to a model in which Notch regulates expression of early temporal factors and early temporal factors regulate Notch activity through translation of Delta mRNA.

There are several strengths of this study. The authors report rigorous measurements and statistical analyses throughout the study. Their conclusions are appropriate for the results. Data mining revealed an important mechanism - that Imp binds Delta mRNA - supporting the model that early temporal factors promote Delta expression, which in turn promotes Notch signaling.

There are also several weaknesses:

- 1. The activation of Notch in NBs by Delta in GMCs was already shown by this group in their Dev 2022 paper, reducing some of the impact of this study.*

In our previous work, we reported that Delta-expressing GMCs transactivate Notch in neuroblasts during the embryonic to larval transition. In the current manuscript, we show that Delta is expressed in GMCs and cortex glia and both sources transactivate Notch in neuroblasts during later developmental stages. This is in agreement with work published by others and while not novel per se, is a necessary first step for understanding which neighboring cell types control Notch pathway activity. During the embryonic to larval transition, glia do not contribute likely because they have not yet grown to ensheath CB NBs and their recently born progeny.

1. *The authors do not explain their current results in context of their prior paper (2022 Dev) until the Discussion, but this would be useful to read in the Introduction. Similarly, it would be good to mention that in the 2022 paper, they find a significant number of wor>Notch RNAi NBs at 2 AHL that are cycling. Are the adult Notch RNAi in this study descended from those NBs at 2 hours ALH in the 2022 study? In other words, how does the early requirement for Notch between 0-72 hours ALH reported in the current study relate to the Notch-depleted NBs identified in the 2022 paper?*

We have now included the following text in the intro: “We recently reported that Notch signaling regulates CB NB quiescence during the embryonic to larval transition (Sood et al., 2022). When Notch is knocked down, some CB NBs continue dividing during this transition. We also reported that Notch activity becomes attenuated in quiescent CB NBs because CB NBs are no longer dividing and producing Delta-expressing GMC daughters for Notch pathway transactivation. Moreover, low Notch is necessary for CB NBs to reactivate from quiescence in response to dietary nutrients (Sood et al., 2022).

Here we report that Notch signaling also regulates neurogenesis termination during pupal stages. When Notch is knocked down, CB NBs maintain early temporal factor expression longer resulting in a delay of late temporal factor expression with prolonged neurogenesis into late pupal stages and early adulthood. This defect in temporal patterning (switching from early to late) occurs after CB NB exit from quiescence suggesting that Notch is required at multiple times throughout development in controlling CB NB proliferation decisions.”

We do not know whether the neuroblasts that fail to enter quiescence are the same that fail to terminate divisions during pupal stages, however there are many more that fail to terminate divisions during pupal stages.

1. *Most of the experiments rely upon continuous depletion of Notch from embryonic stage 8 until adulthood using the wor-GAL4 driver. There is no lineage tracing of this driver and there is no citation about the published expression pattern of this driver. The inclusion of these details is important for a broad audience journal.*

The reference for the driver is included in supplementary data, under the heading “Experimental model:Drosophila melanogaster”. This GAL4 driver is widely used and one of the most accepted in the field.

1. *Most of the experiments utilize a single RNAi transgene for Notch, Delta, Imp, Syp, E93. There are no experiments demonstrating the efficacy of the RNAi lines and no references to prior use and/or efficacy of these lines.*

All RNAi lines used in these studies have been published previously, by our group as well as others and sources for the lines are listed in supplementary data, under the heading “Experimental model:Drosophila melanogaster”. Efficiency of these lines have been verified using antibody labeling (data not shown) and by assaying activity of Notch activity reporters (shown in Fig. 2).

An appraisal: The authors use temperature shifts with Gal80TS to show that Notch is required between 0-72 hours ALH. They show with the use of known markers of the temporal factors and Delta protein trap, that Imp promotes Delta protein expression and the later temporal factors reduce Delta, although the molecular mechanisms are not clearly delineated. Overall, these data support their model that the reduction of Delta expression during larval development leads to a loss of Notch activity.

As noted in the Discussion, this study raises many questions about what Notch does in larval CB NBs. For example, does it inhibit Castor or Imp? Is Notch required in certain neural lineages and not others. These studies will be of interest in the community of developmental neurobiologists.

Reviewer #2 (Public Review):

Embryonic stem cells extensively proliferate to generate the necessary number of cells that are required for organogenesis, and their proliferation must be timely terminated to allow for proper patterning. Thus, timely termination of stem cell proliferation is critical for proper development. Numerous studies have suggested that cell-extrinsic changes in the surrounding niche environment drive the termination of stem cell proliferation. By contrast, cell-intrinsic mechanisms that terminate stem cell proliferation remain poorly understood. Fruit fly larval brain neuroblasts provide an excellent model for mechanistic investigation of intrinsic control of stem cell proliferation due to the wealth of information on molecular marks, gene functions and lineage hierarchy. Sood et al. conducted a genetic screen to identify genes that are required for the termination of neuroblast proliferation in metamorphosis and found that Notch and its ligand Delta contribute to their exit from cell cycle. They showed that knocking down Notch or delta function in larval neuroblasts allows them to persist into adulthood and remain proliferative when no neuroblasts can be detected in wild-type adult brains. By carrying out a well-designed temperature-shift experiment, the authors showed that Notch is required early during larval development to promote timely exit from cell cycle in metamorphosis. The authors went on to show that attenuating Notch signaling prolongs the expression of temporal identity genes castor and seven-up perturbing the switch from Imp to Syp/E93. Finally, they showed that knocking down Imp function or overexpressing E93 can restore the elimination of neuroblasts in Notch/delta mutant brains.

Overall, the experiments are well conceived and executed, and the data are clear. However, the data reported in this study represent incremental progress in improving our mechanistic understanding of the termination of neuroblast proliferation.

We respectfully disagree with this statement. Because Notch signaling is implicated in neurogenesis termination and Notch activity is regulated by GMCs and glia, it strongly suggests that NB proliferation and timing cues are controlled in a non-autonomous manner through direct interactions with NBs and their neighbors. This is in contrast to temporal patterning during embryogenesis which is largely believed to be controlled NB-autonomously. In addition, to our knowledge, no one has yet reported that CB NBs fail to terminate cell divisions on time when Notch activity is reduced during normal development. In fact, reported NB phenotypes associated with Notch loss of function have been surprisingly subtle until now.

Some of the data seem to represent more careful analyses of previously published observations described in the Zacharioudaki et al., Development 2016 paper while others seem to contradict to the results in this study.

The Zacharioudaki et al., Development 2016 paper is terrific. One key difference between our work and theirs, is that we look at Notch pathway knockdown and loss of function phenotypes, whereas in the Zacharioudaki 2016 paper, the authors report phenotypes associated with Notch constitutive activation. It has been known for some time that constitutively active Notch leads to tumorigenic phenotypes particularly in type II lineages. Zacharioudaki and colleagues further determined that some of the classically known

temporal transcription factors were ectopically expressed in these stem cell tumors. Here we show that under normal developmental conditions, Notch pathway activity controls CB NB temporal patterning.

Gaultier et al., Sci. Adv. 2022 suggested that Grainyhead is required for the termination of neuroblast proliferation in a neuroblast tumor model, and grainyhead is a direct target of Notch signaling. Thus, Grainyhead should be a key downstream effector of Notch signaling in terminating castor and seven-up expression. Identical to Notch signaling, Grainyhead is also expressed through larval development. Grainyhead can function as a classical transcription factor as well as a pioneer factor raising the possibility that temporal regulation of neurogenic enhancer accessibility might be at play in allowing Notch signaling in early larval development to set up termination of castor and seven-up expression in metamorphosis. Diving deeper into how dynamic changes in chromatin in neurogenic enhancers affect the termination of neuroblast proliferation will significantly improve our understanding of termination of stem cell proliferation in diverse developing tissue.

Reviewer #3 (Public Review):

In this study, the authors investigate the effects of Notch pathway inactivation on the termination of Drosophila neuroblasts at the end of development. They find that termination is delayed, while temporal patterning progression is slowed down. Forcing temporal patterning progression in a Notch pathway mutant restores the correct timing of neuroblast elimination. Finally, they show that Imp, an early temporal patterning factor promotes Delta expression in neuroblast lineages. This indicates that feedback loops between temporal patterning and lineage-intrinsic Notch activity fine tunes timing of early to late temporal transitions and is important to schedule NB termination at the end of development.

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This is already known for svp and cas and we have now included this information in the discussion. Thank you.

“Whether Notch pathway activity curtails both Cas and Svp or just Cas remains an open question, however it has been reported that both cas and svp are associated with at least one enhancer that is responsive to Notch activity (Zacharioudaki et al., 2016).”

A similar approach has been recently undertaken by the lab of Dr Xin Li, to show that Notch signaling regulates sequential expression of temporal patterning factors in optic lobes neuroblasts (PMID: 36040415), which exhibit a different temporal patterning system than central brain neuroblasts in the present study. As such, the mechanistic insights of the study are limited.

Reviewer #1 (Recommendations For The Authors):

1. There are missing controls

a) Fig. 1F and Fig. 6A - The authors should generate and show images of control clones (FRT19A) stained with the same markers as Notch clones.

Fig. 1F is at 48 hours APF. In control clones, there are no Dpn positive cells present, as stated in the text and therefore no confocal images are shown. Same for Fig. 6A, there are no Dpn positive cells in control clones in the brain at this time, therefore nothing to double label.

1. This result is incorrectly described in the Results

a) P. 5 "Ectopically persisting N RNAi CB NBs expressed the NB transcription factor Deadpan (Dpn), the S-phase indicator pcnaGFP, and were small on average, similar in size to control CB NBs at earlier pupal stages (Fig. 1B,C,E)." The Notch RNAi NBs were larger (not smaller) than controls in Fig. 1E at 30, 48, 72 h APF and in adults.

Thank you for this comment. We have changed the language in the main text as follows:

“Ectopically persisting N RNAi CB NBs (CB NBs at 48 hours APF and beyond) expressed the NB transcription factor Deadpan (Dpn), the S-phase indicator pcnaGFP, and were small on average compared to control CB NBs during earlier developmental stages (L3 control, average diameter 10-15µms) (Fig. 1B,C,E). However, at 30 hours APF when control CB NBs are still present, N RNAi CB NBs were larger on average (Fig. 1B,C,E).”

1. This sentence needs clarification/editing

a) P. 4: "Independent of neurogenesis timing and the mechanism by which CB NB stop divisions, temporal patterning plays a key role". A key role in what?

Thank you again. We have changed the text to the following:

“Independent of neurogenesis timing and the mechanism by which CB NB stop divisions, temporal patterning plays a key role in controlling numbers and types of neurons made within each of the NB lineages (Maurange et al., 2008; Tsuji et al., 2008; Bahrampour et al., 2017; Yang et al., 2017; Pahl et al., 2019).”

1. Some sentences need references or data to support them.

a) P. 9 Please provide a reference to support the statement that Delta is a known Notch target

We have included a reference.

b) P. 9 - please provide a reference or data to support the statement that Delta transcripts decrease over time in larval CB NBs.

This result is shown in Fig. 7B.

1. Fig. 7A - it is difficult to appreciate the purple highlighting.

We have changed the colors as suggested.

Reviewer #2 (Recommendations For The Authors):

1. In Fig. 4C, why does late knockdown of delta lead to ectopic persistence of NBs but late knockdown of Notch has no effect?

This could be due to many things including differences in efficiency of UAS-RNAi lines. The point is that Delta/Notch is required early, but not late. Although some DeltaRNAi CB NBs are still present, the number compared to 48 hours APF is greatly reduced.

1. It is surprising that Delta expression in NBs/GMCs appears to play a more important role in activating Notch signaling in neuroblasts than Delta expression in cortex glia. Please explain how Delta can cell autonomously activate Notch signaling.

We are not proposing that Delta activates Notch cell autonomously, but are proposing that Delta in GMCs transactivates Notch in NBs. After NBs divide Delta is partitioned to GMCs. Quiescent NBs have low to no Notch pathway activity, likely because they are not producing Delta expressing GMC daughters (Sood, 2022).

Please also reconcile the difference in gene expression induced by delta[RNAi] in this study and the delta-mutant allele used in the Zacharioudaki et al study.

We are unsure what the reviewer is asking here and therefore can not reconcile any differences in gene expression between the dIRNAi line and the mutant allele. What gene expression needs to be reconciled? Zacharioudaki is listed as first author on four manuscripts. Which paper is being referred to?

1. In Fig. 2J-L, why does knocking down delta in glia lead to loss of Scrib expression in neuroblasts and their surrounding progeny?

We are not sure if it does or not. We only use Scrib as a membrane marker to identify and locate cells and neuropil regions of interest.

1. The phrase "Notch is active early" is misleading when multiple labs have shown that Notch signaling is active in neuroblasts throughout larval development.

Good point! We have rewritten the statement: “Somewhat paradoxically, we find that early Notch activity is required to terminate CB NB divisions late.”

1. *Neuroblasts that persist into adulthood are "smaller and Dpn-positive/PCNA-GFP-positive". Are they really neuroblasts? Can the authors verify the identity of these "persistent neuroblasts" with other molecular markers as well as functional assessment by inducing lineage clones?*

We have no doubt that these cells are NBs. Because we examine brains over time, these cells can be tracked using the markers, Scrib, Dpn, and pcna. These cells also undergo asymmetric cell division (Refer to Fig. S2F) and express other markers characteristic of CB NBs (mir and insc-not shown). We have made clones and see the same phenotype (ectopic persistence) in both MARCM clones and in “flip-out” clones.

Reviewer #3 (Recommendations For The Authors):

I have a few issues that need to be addressed to reinforce some of the conclusions:

1. *It is unclear whether NBs that persist in late pupal or adult stages have just failed to differentiate or whether they continue to divide, leading to supernumerary progeny (as shown for NBs that are stalled in temporal patterning like in svp mutant NBs (Maurange et al. 2008)). EdU or PH3 staining could be done in adults to clarify this point.*

In this manuscript, we make use of pcna:GFP, a reporter for E2F activity as an indicator of cell proliferation. We certainly observe Dpn positive cells that only weakly express the reporter, suggesting that these cells are not actively dividing or dividing at a reduced rate. However, by far most of the ectopically persisting CB NBs strongly express the reporter and generate pcnaGFP expressing progeny, indicating that these cells are dividing. We have also stained tissues with PH3 and have included an image of a telophase dRNAi expressing CB NB at 48 hours APF (Fig. S2F).

1. *It is unclear whether Notch signaling directly or indirectly regulates temporal transitions. One possibility is that knockdown of Notch signaling decreases cell-cycle speed leading to delayed temporal transitions. The authors should test whether Notch KD affects cell cycle speed using EdU incorporation or PH3 staining. This could be done best using Notch mutant MARCM clones as wt NBs can be used as controls.*

We have quantified the number of PH3 positive CB NBs during wandering L3 stages in control and dRNAi animals. We find that dRNAi CB NBs are indeed proliferating at reduced rates compared to controls. To test whether reduced cell cycle times are causative for termination delay, we expressed a constitutively active form of PI3-kinase in dRNAi animals to drive cell growth and proliferation. We found that CB NBs still ectopically persist (Fig. S2E-G).

We have included the following in the text:

“Defects in timing of temporal transitions could be due to defects in cell cycle progression, although embryonic NBs still transition independent of cell division (Grosskortenhaus et al., 2005). We used PH3 to assay CB NB mitotic activity. In Delta knock down animals, the percentage of PH3 positive CB NBs was reduced compared to control (Fig. S2E). At 48 h APF however, Delta knock down CB NBs were still dividing based on PH3 expression (Fig. S2F). To determine whether CB NBs ectopically persist due to defects in cell cycle rate, we co-expressed dp110 to constitutively activate PI3-kinase in Delta knock down animals. A

significant number of pcnaGFP expressing, Dpn positive CB NBs were still observed, suggesting that defects in cell cycle timing and growth rates alone cannot account for ectopic persistence of CB NBs into later developmental stages and adulthood (Fig. S2G)."

1. *Cas is expressed in NBs either during quiescence and shortly after quiescence. It is possible that the maintenance of Cas in Figure 5D, E is due to NBs that have not re-entered the cell-cycle or have exited quiescence with a strong delay.*

Knockdown of Notch pathway has no effect on CB NB reactivation from developmental quiescence. In fact, low levels of Notch are required for CB NBs to reactivate in response to dietary nutrients (Sood, 2022).

Indeed, the authors have previously shown that Notch signaling is important for NB cell cycle reentry during early larval stages (PMID: 35112131). Are Cas and Svp also maintained in late larval N-/MARCM clones (MARCM clones are made after quiescence exit)?

We have not assayed Cas or Svp expression past 48 hours ALH.

1. *The authors have revisited some previously published RNA-seq data showing that Delta is temporally regulated in NB lineages. This is not clearly shown by the authors that the same is true at the protein level.*

Moreover, they find that mis-expression of late temporal factors or Imp knockdown in early larval brains appear to decrease Delta expression. Such semi-quantitative analysis of gene expression by immunostainings in different conditions can be a bit complicated and not very convincing because variations on intensity levels can be due to slight variations in antibody concentration, or different parameters of image acquisition.

We totally agree, but in this case the difference compared to controls was so readily apparent, that we felt it was not necessary to carry out experiments in clones. All images were acquired with the same confocal settings, experiments were repeated, and we consistently observed the same results. The data shown in Fig. 7D-G is representative.

I suggest that the authors use clonal analysis rather than pan-neuroblast manipulation in order to have internal controls. For example, blocking temporal progression in Svp-RNAi clones (MARCM or Flp-out) and/or svp MARCM clones should lead to maintenance of Imp expression in late larval clones and maintenance of high levels of Delta, which would be easily assessed compared to surrounding NBs.

Minor points:

Fig 5: the sequential expression of Cas and Svp expression in larval NBs was first described by Maurange et al. 2008. Please cite appropriately.

We have now added the requested citation to the following:

"Over time, the percentage of Cas expressing CB NBs declined, while Svp expressing CB NBs modestly increased (Fig. 5B). Less than 1% of CB NBs co-expressed Cas and Svp at any stage and expression of both factors was absent by 48 hours ALH (Fig. 5B,C). This is consistent with work published previously (Isshiki et al., 2001; Tsuji et al., 2008; Chai et al., 2013; Maurange et al., 2008; Ren et al., 2017; Syed et al., 2017)."

Fig 6A: Please indicate which immunostainings are shown in the overlay panels.

Good catch! We have modified the figure.

P9: "Delta co-immunoprecipitated with Imp.": Add "Delta mRNA co-immunoprecipitated with Imp in RIP-seq experiments" Otherwise, it suggests that you are talking about the protein.

Done

The scheme in Figure 7H is rather complicated to understand. In my opinion, it does not clearly convey the idea that Notch signaling favors the Imp-to-Syp transition.

We have made a new model figure.